TSK: A Trustworthy Semantic Keypoint Detector for Remote Sensing Images

Jingyi Cao, *Student Member, IEEE*, Yanan You, *Member, IEEE*, Chao Li, and Jun Liu, *Member, IEEE*

***Abstract*— Keypoint detection aims to automatically locate the most significant and informative points in remote sensing images (RSIs), which directly affects the accuracy of matching and registration. In contrast to the handcrafted keypoint detectors that heavily rely on the morphological gradient of corner, line, and ridge, the learning-based detectors emphasize obtaining reliable keypoints from deep features. However, the limited accuracy of semantics undermines the reliability of keypoints, especially in challenging scenarios characterized by repeated textures and boundaries. Therefore, a novel trustworthy semantic keypoint (TSK) detector is proposed for RSIs. It utilizes a lightweight multiscale feature extraction and fusion network, along with a saliency keypoint localization mechanism, to facilitate keypoint detection. Notably, the TSK detector employed explicit semantics, which is refined with multiple learning strategies about repeatability and representability across the multigranularity reasoning spaces, namely, pixel window, neighbor window, and existence entity. Finally, several metrics about repeatability, matching, and registration are used to evaluate the performance of the TSK detector and other competitive methods. Four RSI datasets, including MICGE, HRSCD, OSCD, and SZTAKI, are used to verify performances. TSK detector achieves competitive performance against existing methods.**

***Index Terms*— Feature descriptor, feature interpretability, image registration, keypoint detection, remote sensing.**

I. Introduction

KEYPOINT detection is a fundamental problem in the image processing field. The pairwise keypoints contribute to various tasks such as visual localization [1], 3-D reconstruction [2], image matching [3], [4], [5], and relative radiometric normalization [6], [7]. Especially, for the registration task of the multitemporal remote sensing images (RSIs), a validity keypoint detector significantly impacts the registration result.

Relying on the gradient, intensity, and curvature, the handcrafted detectors adopt the predefined rules for keypoint detection [10], [11], [12], [13]. In fact, there is no worldwide agreed-upon definition for keypoint. Blobs, edges, corners, and even conspicuous spots are frequently handcrafted keypoint forms for the image processing field. However, owing to spatial gray color value fluctuation caused by the random noise and repetitive textures, many unexplained keypoints are detected, which are usually useless for subsequent tasks. Several works attempt to introduce deep semantics into the handcrafted keypoint detection, including histogram [14] and encoded color vector [15]. Besides, spatial continuity is also of interest to researchers. Edges, ridges, and blocks are effective auxiliary information [16], [17]. Whatever employing the chain code descriptors [18] and moment invariants to extract boundary and contour [19] or using threshold and panoramic segmentation [20] to separate salient regions, regional closure is core to ensure the available points. Nevertheless, the internal points are disregarded, despite they are significant for subsequent matching tasks, and the time expenditure cannot be overlooked.

In fact, rather than adopting designated handcrafted patterns, describing keypoints from the semantic level achieves better antinoise interference for keypoint extraction. In recent years, the convolutional neural network (CNN) has shown its ability to local feature expression and spatial feature mining. Consistent with the improvement idea of semantics and spatial correlation complementarity as the handcrafted schemes, the cascaded convolution helps the network to abstract semantically rich features with a larger spatial receptive field [21]. Compared with the shallow morphological features, the entity semantics implied in the deep feature display better robustness. As a milestone, SuperPoint [22] utilizes the handcrafted keypoints as the pseudotags for the supervised learning. Nevertheless, imprecise handcrafted keypoints have raised suspicion among researchers regarding the ability of unreliable labels to produce trustworthy keypoints. In fact, it is biased to define keypoint as positions or linear regions with notable feature changes. A more robust way is to extract explicit features according to the attributes of the keypoint and neighbor relationships. Following the guidance of repeatability, the semantic distance of the descriptors to the point pair on multitemporal images provides an optimization target for network learning. Under different spatial granularities and information spaces, UnSuperPoint [23], D2-Net [8], R2D2 [9], and LIFT [24] are the representatives that carry out repeatability determination for keypoint learning. The generated points offer advantages for the subsequent matching, as they are not restricted by handcrafted keypoint setting and are entirely derived from the cognition of computer model [25].

However, the current semantic methods do not guarantee the trustworthiness of the extracted keypoints. It is confirmed in Fig. 1 that, whether the competitive learning-based methods R2D2 or D2-Net, inaccurate keypoints are visible in areas without explicit semantics, such as sea and lake, which are considered unstable and noisy [indicated by red circles in

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The authors are with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: caojingyi@bupt.edu.cn; youyanan@bupt.edu.cn; chaoli1998@163.com; liujun@bupt.edu.cn).

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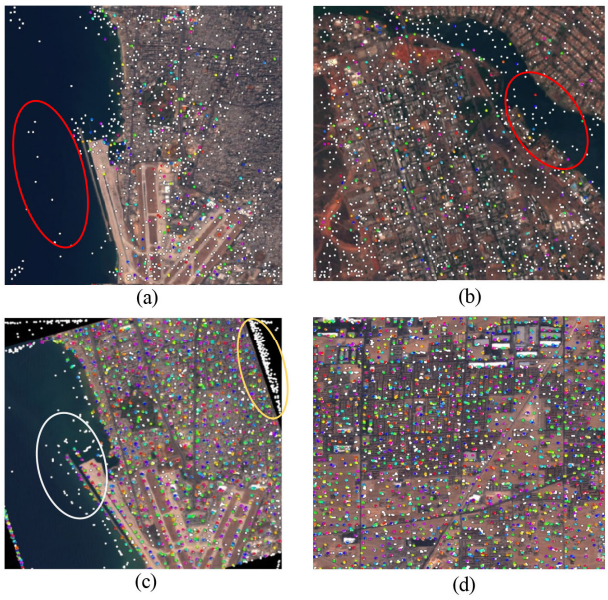


Fig. 1. Examples of keypoint results. (a) and (b) are generated by D2-Net [8]. (c) and (d) are generated by R2D2 [9]. The white points indicate the absence of the correspondings, while other colors represent their presence. Regions containing keypoints without explicit semantics are highlighted in circles.

Fig. 1(a) and (b)]. Furthermore, keypoints may be erroneously detected at color block junctions on a flat surface [indicated by the white circle in Fig. 1(c)]. Moreover, despite plentiful enough keypoints distributed across the image, they may not be precisely aligned with texture boundaries, as depicted in Fig. 1(d). Apart from the imprecise feature expression caused by the scale difference of the receptive field between descriptor and pixel, the most important reason is that the semantics extracted by these methods are not reliable. Trained with unclear keypoint definitions, descriptors with weak discrimination are bound to generate ambiguous keypoints.

For obtaining accurate and meaningful keypoints, a novel trustworthy semantic keypoint (TSK) detector is proposed. The attributes of keypoints, including repeatability and representativeness, provide us with inspiration for strategy-making. On this basis, effective feature refinement is conducted within the multigranularity reasoning spaces of keypoints. Specifically, we employ feature heterogeneity to assess whether the observed window contains keypoints. It quantifies the nonuniform distribution of information within the window, calculated as the sum-average of the information entropy to pixels. A lower value indicates a higher level of heterogeneity, representing a greater likelihood of keypoints being present. However, heterogeneity is insufficient for precise localization; neighbor points and existence entities are incorporated to enhance the reliability of local spatial cognition. Corresponding to different spatial granularities, the existence reinforcement module, the trustworthy keypoint induction module, and the descriptor alignment module are proposed to guarantee the double trustworthiness of both keypoint position and descriptors semantics. "Detect-and-refine" is completed to give the descriptor explicit instructions for keypoints. In summary, our work encompasses the following contributions.

1) Based on the encoder–decoder structure, the TSK detector is proposed. It is unsupervised and does not require any manual annotation.
2) A lightweight multiscale feature extraction and fusion network and a saliency keypoint location mechanism are proposed. Owing to the consideration of uniformly arranged multigranularity windows, it effectively avoids the uneven global keypoint distribution caused by the weak feature response of dark areas.
3) In terms of the spatial granularity from detail to local, a novel paradigm for describing keypoint is proposed by constructing multigranularity keypoint existing spaces, namely, pixel window, neighbor window, and existence entity, respectively. On this basis, several feature refinement strategies are set according to repeatability and region representability.
4) Four multitemporal RSI datasets, including MICGE [3], OSCD [26], HRSCD [27], and SZTAKI [28], are applied for performance evaluation. The trustworthiness of keypoints generated by TSK and other detectors is verified by repeatability together with the downstream application matching and registration.

To provide a clear structure, our work is organized into the following sections. Section II reviews the development of keypoint detection. Section III introduces the details of the TSK detector according to the multigranularity existing spaces. Section IV provides implementation and experimental details on the multitemporal RSI datasets. Section V ultimately reaches a conclusion about our work.

II. Related Work

A. Handcrafted Detector

The handcrafted schemes regard keypoint detection as a process of identifying spatial extreme points (e.g., corners and blobs). They detect keypoints according to the variation in pixel gray values and the intensity changes within the observation window. Harris [10], gradient-based Hessian matrix [speeded-up robust features (SURF)] [29], and threshold decision in window [features from accelerated segment test (FAST)] [12] are feasible solutions. However, some meaningless keypoints are possibly extracted due to the gray-scale fluctuation caused by noise. To ameliorate noisy sensitivity, the visual pyramid and region analysis are introduced. Methods include scale invariant feature transform (SIFT) [30] with Gaussian difference pyramids, oriented FAST and rotated BRIEF (ORB) [13] that refines FAST with image pyramids, and binary robust invariant scalable keypoints (BRISK) [31] utilizing Gaussian filtering followed by subpixel interpolation. The primary objective is to scale the low-level features for filtering out interferences without significant semantics whether intentionally or unintentionally.

In recent years, complementing spatial distribution and local semantic information has become a research hotspot for handcrafted detectors, including, but not limited to, histogram, edge, and texture features that are used to optimize eigenvector representation. For instance, it is available to describe features with color histograms or orientation gradient histograms [14] or add descriptive information such as saliency texture and

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edge into the original features [32]. Temporally invariant learned detector (TILDE) [15] appends time, lightness and chromaticity (LUV) color, and gradient information to the low-level features extracted from SIFT, followed by keypoints classification on the generalized hinging hyperplanes (score maps). Especially, the recent work HarrisZ+ [33] integrates multiple keypoint detectors with other modern components, achieving keypoints with richer visual properties. However, defining keypoint as spatial extreme points is not comprehensive. Limited feature recognition and noise robustness remain the drawbacks.

B. Learning-Based Detectors

Neural networks have revolutionized feature extraction by generating features with deep and rich semantics, which has eliminated the need for manual feature engineering and strongly propelled the advancement of learning-based methods. As a representative for fusing between handcrafted and neural networks, Key.Net [21] operates feature aggregation as Harris in feature map, and D2D [34] mines pixel differences using gradient difference. Their better performances prove that the handcrafted strategies not only adapt to deep features extracted from the neural network but also make full use of the richer semantics inside. Moreover, the larger receptive field and semantic cognition contribute to suppressing noise. For further developing the adaptability of learning-based detectors, two optimization methods have emerged, namely, learning strategy and feature description [35], [36], [37], [38].

1) *Learning Strategy:* Fundamentally, MagicPoint [39] and SuperPoint [22] transformed the task of keypoint extraction into regression, taking points extracted from the conventional methods as supervision label. However, standard supervised training methods limit the network's perception of keypoints. Instead, R2D2 [9] and UnSuperPoint [23] build a new mode for self-supervision learning based on repeatability. They construct the training samples with the help of rotation augmentation and then minimize the semantic distance of feature vectors to the paired points within the multiperspective [40]. As another unsupervised metric, reinforcement learning [41] is also adopted to enable the model to learn keypoints adaptively.
2) *Feature Description:* Dealing with the multiscale and direction inputs, feature pyramid, dilated convolution, and rotational equivalent structure [42], [43], [44] are effective solutions for feature description. Moreover, in addition to point features, the local descriptor, which is the unified feature vector containing point features and neighbor features, shows advantages for keypoint cognition and location. As representatives, D2-Net [8] and ASLFeat [43] carry out feature description in sparse spatial grids (such as 8×8) and then obtain the corresponding keypoint (CKpt) in each grid based on the local features, which can be summarized as "detect-and-describe." Furthermore, combining spatial relationships and repeatability, some works construct keypoints from the perspective of patch matching, in which the pixel with the highest feature response within each patch is selected as keypoint [45]. Examples are LF-Net [46], LIFT [24], and LoFTR [47]. Experiments prove that considering neighborhood spatial correlation [48] and applying transformer-related attention mechanism [47] bring competitive matching results even in distorted cases. However, most of their keypoints are offset from the keypoint detected by the handcrafted methods due to the ill-defined morphology of the keypoints.

For the learning-based detectors, keypoints are extracted according to the semantics contained in the deep features. Only on the premise of semantic credibility, the trustworthy keypoints can be guaranteed. However, the current approach is not available for reliable semantic generation. The reason is that there is no explicit definition for keypoint in terms of the existence or representativeness at the semantic level. Therefore, our dedication lies in describing the repeatability and representation of keypoints within multigranularity reasoning spaces. Specifically, the decisive role of the relative relationship between the location and features of neighboring keypoints is emphasized. Besides, considering that interference may be caused by applying a unified description for both keypoint detection and patch matching [49], our work emphasizes specific keypoint descriptors rather than the universal one.

III. Methodology

A. Overview

Following the "detect-and-describe" mode, keypoint detection can be summarized into the following model:

$$p_i := \underset{A_i \in I}{\operatorname{argmax}}(S) \tag{1}$$

where image I can be partitioned into multiple subspaces corresponding to spatial granularity and the keypoint set P consists of keypoints detected in each subspace A_i, determined by the feature response in the decoded score map S. In objective terms, there are various choices available for the local space A_i. For example, in FAST [12], A_i is set as the circular space with a radius of r. Emphasizing local features, the following available multigranularity spaces are listed.

1) *Pixel Window:* The minimum grid cell for keypoint determination (8 in our work)

$$A_i \subseteq \mathbb{R}^{8\times 8}. \tag{2}$$

2) *Neighbor Window:* Grids of 3×3, encompassing both the central keypoint and neighboring features. Each grid corresponds to the pixel window

$$\begin{aligned} A_i \subseteq & \begin{bmatrix} b_{i-1,j-1} & b_{i-1,j} & b_{i-1,j+1} \\ b_{i,j-1} & b_{i,j} & b_{i,j+1} \\ b_{i+1,j-1} & b_{i+1,j} & b_{i+1,j+1} \end{bmatrix} \\ \text{s.t. } & b_{i,j} \subseteq \mathbb{R}^{8\times 8} \\ & p_i \in b_{i,j}. \end{aligned} \tag{3}$$

3) *Existence Entity:* Space derived from keypoint p_i and its highly correlated neighborhood keypoint set R_i with algorithm $\mathcal{G}$. Although its spatial center may not necessarily be the concerned keypoint, this space is likely to coincide with the entity where keypoint exists

$$A_i \subseteq \mathcal{G}(p_i, R_i). \tag{4}$$

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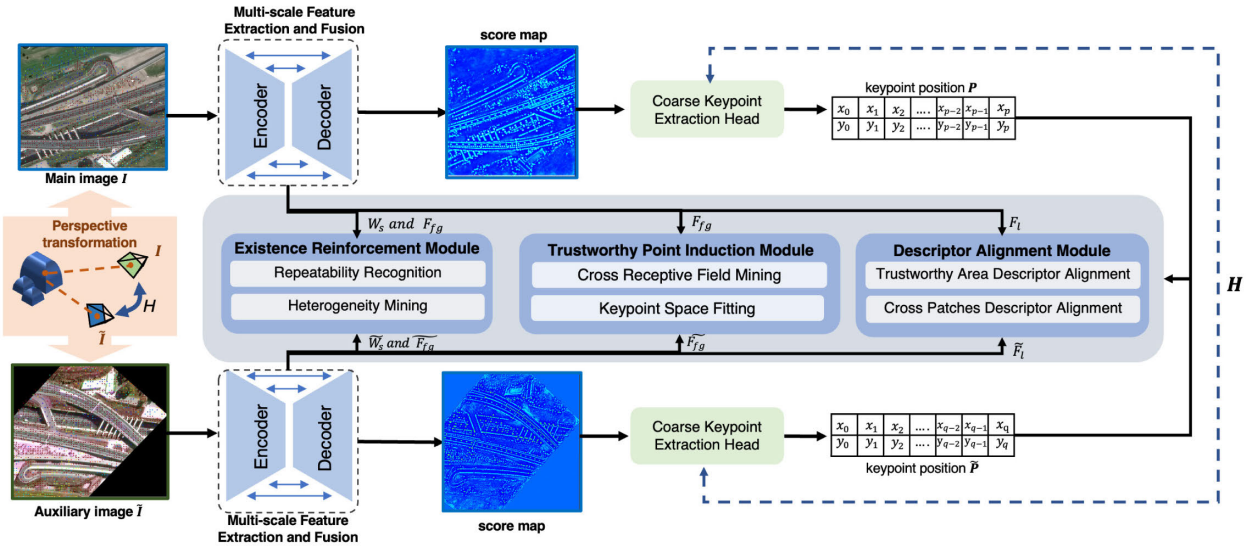


Fig. 2. Overall framework of the proposed TSK detector. The symmetric dual-branch network utilizes a codec structure with a multiscale feature fusion module to extract keypoints and corresponding descriptors from the main image I and the auxiliary image $\widetilde{I}$. During training, the descriptor is refined based on the relationship between features and keypoints through the existence reinforcement module, the trustworthy point induction module, and the descriptor alignment module.

In fact, regardless of the spatial granularity, keypoints must adhere to constraints of representation and repeatability. More specifically, we take the descriptors corresponding to the pixel positions of p_i as $f(p_i)$ and take their corresponding local spatial features as $F(A_i) = \sum_{p \in A_i} f(p)$.

1) *Regional Representation:* Keypoint is available to represent the local features of its belonging space. $\Re$ can be any function for feature consistency measure

$$\begin{aligned} &\forall q_j \in Q_i \subseteq A_i \backslash p_i, \quad F(A_i) \ncong 0 \\ &\Re(f(p_i), F(A_i)) \gg \Re(f(q_j), F(A_i)). \end{aligned} \tag{5}$$

2) *Repeatability:* For multitemporal images I and $\widetilde{I}$ that undergo a spatial transformation $H \in \mathbb{R}^{3\times3}$, the keypoint pair correspondings to the same feature (or entity) should be extracted simultaneously

$$\widetilde{f}(Hp_i) = f(p_i). \tag{6}$$

In order to facilitate the learning of keypoint features and promote the trustworthiness of the generated keypoints, diverse learning rules are established corresponding to the multigranularity spaces from fine to coarse. Fig. 2 illustrates the architecture of the TSK detector, which comprises a symmetrical dual-branch network for processing main image I and auxiliary image $\widetilde{I}$ during model training. It employs a codec structure with a multiscale feature fusion module to extract keypoints and their corresponding descriptors. The homography matrix (H) of the multitemporal images is known or randomly generated, acting as the self-supervised guidance by perspective transformation. Furthermore, based on detection results, three feature refinement modules are proposed according to three levels of granularity: pixel window, neighbor window, and existence entity. Correspondingly, the existence reinforcement module, the trustworthy point induction module, and the descriptor alignment module are utilized.

B. Description and Keypoint Detection

1) Multiscale Feature Extraction and Fusion: Suppose that the main image $I \in \mathbb{R}^{H\times W\times C}$ and the auxiliary image $\widetilde{I} \in \mathbb{R}^{\widetilde{H}\times\widetilde{W}\times C}$. Notably, subsequent operations on the auxiliary image are symbolized with $\widetilde{(\cdot)}$. The paired images are synchronously input into the dual-branch network and processed with the multiscale feature extraction and fusion module (MsFEF).

As shown in Fig. 3, the MsFEF module employs an encoder–decoder structure with VGG16 as the backbone. During coding, convolution, exponential linear unit (ELU) activation, and pooling are performed repeatedly (no pooling for the last EnBlock). Regarding the decoder, to obtain the keypoint response feature with the same size as the input, the learning-based deconvolution operation is employed, accompanied by instance normalization and ELU activation. Besides, inspired by Ronneberger et al. [50], the skip-connect is applied to enable the decoder features to incorporate the multiscale information $F_{ms}^i \subseteq \{F_{ms}^1, F_{ms}^2, F_{ms}^3, F_{ms}^4\}$ obtained during the encoder stage. As a supplement, concatenation $\oplus$ and convolution operations are added in the intermediate process for organizing the feature channel as follows:

$$F_d^{i-1} = \begin{cases} \text{conv}\left(F_{ms}^i \oplus \text{Tconv}\left(F_d^i\right)\right), & 0 < i \leq 3 \\ \text{conv}\left(\text{Tconv}\left(F_{ms}^i\right)\right), & i = 4. \end{cases} \tag{7}$$

For the accurate location, the deformable convolution is introduced to the last layer in the decode stage. It optimizes the details of the decoded feature and provides position correction ability. Finally, score map $S \in \mathbb{R}^{H\times W\times 1}$ is generated, each position of which represents the keypoint probability.

In addition, deep semantics of the encoder features determine the credibility of the score map. Therefore, we propose a refined process for descriptor generation. Specifically, after the concatenation of the scaled multiscale features F_{ms}^i, the descriptors with different granularity are generated by

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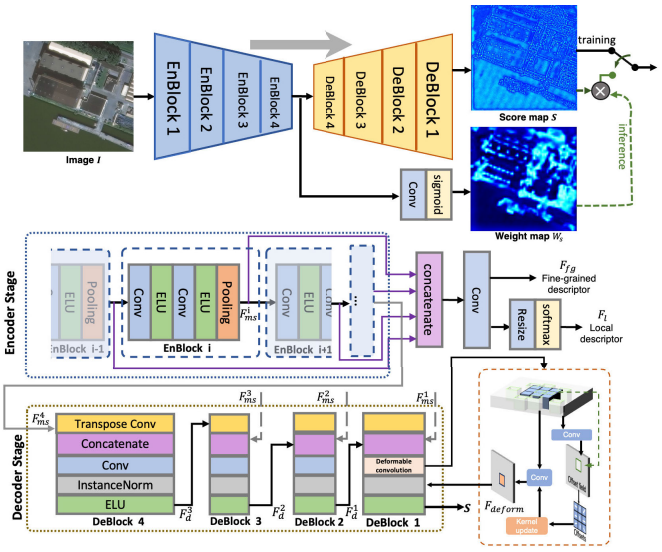


Fig. 3. Implementation details of the MsFEF module.

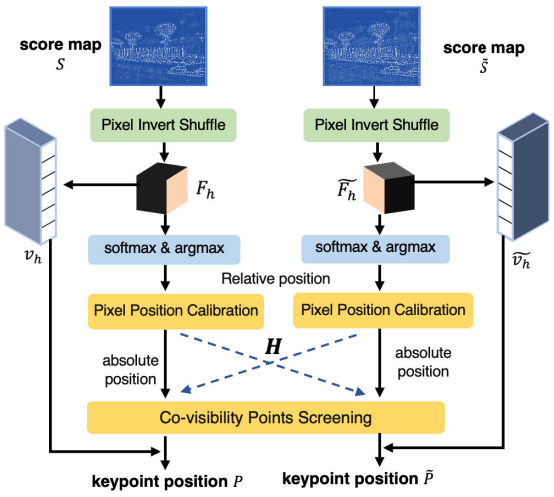


Fig. 4. Framework of the coarse keypoint extraction head.

convolution and bilinear interpolation. The descriptors consist of: 1) fine-grained descriptors F_{fg} whose size is consistent with the input and 2) local descriptor F_l that aggregates local features with scaling and activation operation. In addition, we designed a tensor W_s for delineating regional reliability, which is generated with the last encoder feature $\mathrm{F}_{ms}^4 \in \mathbb{R}^{(W/8)\times(H/8)\times 64}$. After convolution, the sigmoid operation is performed to ensure $W_s \in [0, 1.0]$.

2) Coarse Keypoint Regression Head: The framework is shown in Fig. 4. Consistent with other learning-based detectors, we utilize the fixed 8×8 grid (i.e., pixel window) to produce sparse and uniformly distributed keypoints across the image. Following a sequential order from top left to bottom right, each pixel window is numbered according to the function $\psi(\cdot, \cdot)$ as (8a), and the inverse process is given in (8b)

$$\varphi = \psi(X_m, Y_n) = 8 \cdot X_m + Y_n \tag{8a}$$

$$\overline{X}_\varphi, \overline{Y}_\varphi = \left\lfloor \frac{\varphi}{8} \right\rfloor, \quad \text{mod}(\varphi, 8) \tag{8b}$$

where X_m and Y_n represent the horizontal and vertical indices of each pixel window for S and the window sequence order $\varphi \in [0, (WH/64) - 1]$.

In fact, max pooling has the ability to return the index of position with the maximum feature response within the local window, whereas it directly cuts off the possibility of analyzing regional information. Innovated from pixel shuffle [51], we put forward to apply pixel invert shuffle for deducing keypoint position. It establishes a correspondence between channels and positions, specifically mapping the position within the 8×8 windows to the corresponding $C = 64$ channels. Based on the score map S, we denote the transformed tensor as $F_h \in \mathbb{R}^{(HW/(8\times 8))\times 64}$, where each pixel corresponds to pixel windows. Subsequently, to acquire the relative index of the keypoint within each corresponding window, for each 8×8 window numbered with φ, its feature vector v_h^φ is calculated by channel-level softmax and argmax, as shown in (9a) and (9b), where $v_h^\varphi \in \mathbb{R}^{1\times 64}$ and ind $\in [0, 63]$

$$v_h^\varphi = \text{softmax}(F_h(\varphi)) = \frac{e^{F_h(8X_m+Y_n)}}{\sum_{i=0}^{C} e^{F_h^i(8X_m+Y_n)}} \tag{9a}$$

$$\text{ind}(\varphi) = \underset{A_\varphi \in S}{\text{argmax}}\left(v_h^\varphi\right). \tag{9b}$$

Afterward, taking keypoint $p_\varphi = (x_\varphi, y_\varphi)$ as an example, for obtaining its position, a series of inverse calculations is conducted as formulated in the following:

$$x_\varphi = 8\overline{X}_\varphi + \left\lfloor \frac{\text{ind}(\varphi)}{8} \right\rfloor \tag{10a}$$

$$y_\varphi = 8\overline{Y}_\varphi + \text{mod}(\text{ind}(\varphi), 8). \tag{10b}$$

C. Existence Reinforcement Module

1) Repeatability Recognition: With the known homography matrix H, the mapping between the CKpts (x, y) and (x', y') in multiperspective views can be calculated with function Φ, and $(\cdot)^{-1}$ represents the inverse calculation

$$\left[x', y', z'\right] = H[x, y, 1]^T \tag{11a}$$

$$\Phi\left(H, (x, y)^T\right) = \left[\frac{x'}{z'}, \frac{y'}{z'}\right]^T. \tag{11b}$$

Following the pairing principle that the second element in a tuple belongs to the same pixel windows of the first element, two sets corresponding to coordinate systems of the multitemporal images are established, namely, $U = \{\ldots, (p_j, \Phi^{-1}\widetilde{p_k}), \ldots\}$ and $V = \{\ldots, (\widetilde{p_n}, \Phi p_m), \ldots\}$. Similarly, the score map S and $\widetilde{S}$ are paired up as $(S, \Phi^{-1}\widetilde{S})$ and $(\Phi S, \widetilde{S})$ for quick addressing. However, there are some regions that cannot be displayed in two images simultaneously, owing to the difference in perspective. Therefore, covisibility points' screening is performed to delete the keypoint whose corresponding point exceeds the image boundary. As illustrated in (12), for $p \in P$ and $\widetilde{p} \in \widetilde{P}$, the points satisfied the below conditions are retained

$$\begin{aligned} &0 \le \left(\Phi^{-1}\tilde{p}\right)_x < H, 0 \le \left(\Phi^{-1}\tilde{p}\right)_y < W \\ &0 \le (\Phi p)_x < \widetilde{H}, 0 \le (\Phi p)_y < \widetilde{W} \end{aligned} \tag{12}$$

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where $(\cdot)_x$ and $(\cdot)_y$ represent the vertical and horizontal coordinates of points, respectively.

As repeatability requested, the paired keypoints should obey the consistency in both positions (after coordinate alignment) and semantics. Therefore, by incorporating the cross entropy between the features of point pairs into the loss function, L_{Rep} is calculated as in (13b)

$$\Gamma(p(x,y)) = v_h^{\psi(x,y)} \quad \widetilde{\Gamma}(p(x,y)) = \widetilde{v_h}^{\psi(x,y)} \tag{13a}$$

$$\begin{aligned} L_{\text{Rep}} &= \sum (CE_U + CE_V) \\ &= -\sum_{p_i,p_j\in U} \left[\Gamma(p_i)\log\widetilde{\Gamma}(\Phi p_j) + \widetilde{\Gamma}(\Phi p_j)\log\Gamma(p_i)\right] \\ &\quad - \sum_{p_i,p_j\in V} \Big[\widetilde{\Gamma}(p_i)\log\Gamma(\Phi^{-1}p_j) \\ &\qquad + \Gamma(\Phi^{-1}p_j)\log\widetilde{\Gamma}(p_i)\Big]. \end{aligned} \tag{13b}$$

Despite the 8×8 pixel window, it is also necessary to take into account the feature repeatability at the image level. As mentioned before, the regional reliability tensor W_s serves as a confident mask for evaluating the semantic validity. Derived from the concatenation features constructed for keypoint detection, it is expected that regions, where keypoints are clustered, should exhibit high feature responses with spatially contiguous. Hence, the regional reliability distance loss $L_{\text{RegionRel}}$ is designed as follows:

$$L_{\text{RegionRel}} = \left\| W_s - \Phi^{-1} W_{\tilde{s}} \right\|_2. \tag{14}$$

Taking regional reliability into account of feature optimization, W_s and $W_{\tilde{s}}$ are supplemented to (13b). It is believed that only the regions that have explicit semantics in both W_s and $W_{\tilde{s}}$ are considered trustworthy areas for keypoint learning. As illustrated in (15), the updated $W^A \in \mathbb{R}^{(W/8)\times(H/8)\times 1}$ is calculated by the elementwise multiplication

$$W^A = W_s \cdot \left(\Phi^{-1} W_{\tilde{s}}\right). \tag{15}$$

Furthermore, we rewritten L_{rep} as the following formulation:

$$L_{\text{Rep}} = \frac{1}{\sum_{A_i\in I} W_i^A} \sum_{a_i\in A} (CE_I + CE_{\tilde{I}}) \cdot W_i^A. \tag{16}$$

2) Heterogeneity Mining: In small-scale local windows (e.g., 8×8), the differences between elements are not significant, posing a challenge in determining which is the most representative internal position. Further deriving the representation formula of (5), it can be concluded that the reliable inference of keypoint representativeness is achieved under the condition of minimized correlation between keypoints and surrounding pixels, namely, $\Re(f(p_i), f(q_j)) \cong 0$. Therefore, we shift our approach toward maximizing spatial differences, thereby achieving local spatial heterogeneity. As verified in our study [52], the feature response distribution within the observed region provides favorable evidence for keypoint existence. Therefore, the degree of feature heterogeneity is adopted to represent the intensity of keypoint existence. Especially, concerning the 8×8 pixel window, the heterogeneity of each pixel window A_i is defined into $H(A_i)$, as illustrated in the following:

$$H(A_i) = -\sum_{j=1}^{n} v_h^{ij} \log\left(v_h^{ij}\right) \tag{17}$$

where n is the amount of channels in v_h^i; for pixel windows of 8×8, we set $n = 64$.

Our expectation is that credible keypoints appear when there is high heterogeneity, indicating unevenly distributed features within the observed window. Coincidentally, it is consistent with the region's reliable tensor W_s, presenting a high possibility in trustworthy areas. Therefore, maintaining the synergy of "heterogeneity—spatial credibility," the heterogeneity loss is defined in (18). With the aid of regional reliability, the sporadic high-energy flicker noise is greatly suppressed

$$\begin{aligned} &L_{\text{Heter}} \\ &= \frac{1}{\sum_{A_i\in I} W_i^A}\left[\sum_{A_i\in I} W_i^A \left(H(A_i) + H\left(\tilde{A}_i\right)\right)\right] \\ &\quad - \frac{1}{\sum_{A_i\in I}\left(1 - W_i^A\right)}\left[\sum_{A_i\in I}\left(1 - W_i^A\right)\left(H(A_i) + H\left(\tilde{A}_i\right)\right)\right]. \end{aligned} \tag{18}$$

Conclusively, the loss function about pixel window is defined by the following equation:

$$L_{\text{Pixel}} = L_{\text{Rep}} + L_{\text{RegionRel}} + L_{\text{Heter}}. \tag{19}$$

D. Trustworthy Keypoint Induction Module

Owing to the limited observation coverage, feature restriction based on pixel window can only discover points rather than the trusted keypoints. Therefore, focusing on the neighbor window, two components of representativeness are considered: the cross-receptive field mining component (CRFM) and the keypoint space fitting (KSF) component. Note that the repeatability of keypoints will be specifically addressed in the existing entity, hence not mentioned here.

1) Cross-Receptive Field Mining Component: Avoiding the errors caused by the cross-scale locating, the fine-grained features F_{fg}, which is of the same scale as the original image, are used to describe keypoints. Specifically, for each keypoint p_i, the centered clipped features $f_c^i \in \mathbb{R}^{8\times8\times C}$ are generated from F_{fg}. In addition, keypoints are sampled with a ratio of 0.3 for time efficiency, namely, $p_i \in P_s \subset P$ or $\widetilde{p_i} \in \widetilde{P_s} \subset \widetilde{P}$. Then, through the fusion function $\mathfrak{F}(\cdot,\cdot)$, the detail features f_c^i is fused with the neighbor local features $f_n^i \in \mathbb{R}^{(8\times3)\times(8\times3)\times C}$, which is also clipped centered on the CKpts. Finally, the fused features are compressed into a 3×3 tensor X^i by bilinear interpolation $\mathfrak{B}$, remining the relationship between the central and neighborhood. The implementation detail of the CRFM component is illustrated in Fig. 5, and the regional features are obtained by the following equations:

$$X^i = \mathfrak{B}\left(\mathfrak{F}\left(f_c^i, f_n^i\right), (3,3)\right) \tag{20a}$$

$$X_c^i = X_{(1,1)}^i \tag{20b}$$

$$X_n^i = \left\{X_n^{ij} \middle| X_n^{ij} \in X^i / X_c^i\right\} \tag{20c}$$

$$X_d^i = \mathcal{AP}\left(X^i\right) \tag{20d}$$

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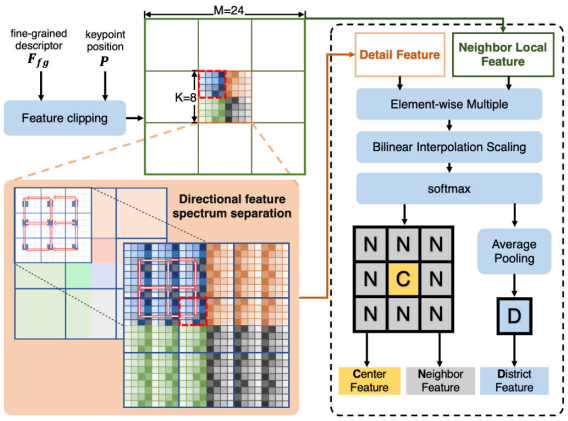


Fig. 5. Implementation details of the CRFM.

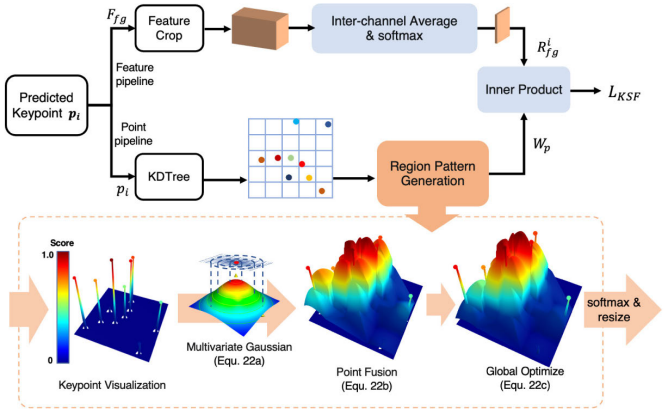


Fig. 6. Implementation of the KSF component.

where the central features are X_c^i, neighbor features are X_n^i, and the district features are X_d^i. $\mathcal{AP}$ represents the average pooling.

Equation (5) demonstrates that the high similarity between central and global features confirms the representativeness of keypoints. We use cosine similarity for evaluating correlations, where a value of 1 indicates feature consistency. To prevent global averaging or zero padding, we introduce a game setting to enforce a high degree of differentiation between the central feature and its neighborhood. This is achieved by calculating the cosine similarity between the center and the surrounding $K = 8$ regions. The CRFM-related loss is given as follows:

$$L_{\text{CRFM}} = \frac{1}{M} \sum_{p_i \in P_s \bigcup \widetilde{p}_i \in \widetilde{P}_s}^{M} \left[1 - \Re\left(X_c^i, X_d^i\right) + \frac{1}{K} \sum_{j}^{K} \Re\left(X_c^i, X_n^{ij}\right) \right]. \quad (21)$$

It needs to be stressed that for fusing detail features with the neighboring local features, the directional feature spectrum separation (DFSS) operation is employed as $\mathfrak{F}$. It is observed that in the keypoint detection task, the learned convolution kernel tends to focus on the high-frequency details, with each layer acting as a high-frequency extraction operator. However, the ignored low- and medium-frequency features are not meaningless [18], [19]. Therefore, we desire to produce features about various granularities, that is, to obtain operators focused on different frequencies. As we all know, convolving with a periodic pulse signal corresponds to spectrum sampling in the frequency domain and replication in the spatial domain. Therefore, we propose using spatial feature copying to achieve frequency sampling while adjusting the feature map to the desired size. This operation is preferred over simple scaling, owing to it not introducing any uncertain interpolation information. Note that, rather than uniformly spreading the 8×8 features across all eight neighboring domains, we divide the 8×8 block into four bins and perform feature extension with small strides within their corresponding corner spaces, as shown in the bottom left of Fig. 5. There are two reasons: 1) feature replication across the entire window disrupts feature continuity and 2) long-stride replication is equivalent to dense sampling in the frequency domain, which is unfavorable for maintaining a balanced spectrum. The effectiveness of DFSS will be proved in the experiment.

2) KSF Component: For corner keypoints extracted from the conventional detector, keypoints are often located in the line region connected to the corner, aside from the inflection point with the most pronounced curvature. The relationship between the adjacent keypoints in terms of geometric positions and feature intensity assists in reasoning the attributes of geometric elements. Therefore, we construct region patterns based on the known location and confidence of the extracted keypoints and utilize them to achieve a close fit for the corresponding features.

Similar to CRFM, we take the clipped features f_c^i as input. Then, interchannel average and softmax are performed for generating the morphological tensor $R_{\text{fg}}^i \in \mathbb{R}^{8\times8\times1}$, as the upper branch in Fig. 6. Besides, region patterns come from the region pattern generation components, as shown in the lower part of Fig. 6, where keypoint confidence is represented by color following the jet mapping. With all sample keypoints P_s (or $\widetilde{P}_s$), the k-dimensional tree (KDTree) is adopted to search the nearest neighborhood keypoints $P_i^{\text{neighbor}} = \{p_i^0, \ldots, p_i^m\}$ of the extracted point p_i within the search space $(l, l) = (5 \times 8, 5 \times 8)$. Since the keypoints are discrete, the multivariate Gauss tensor $\mathfrak{G}$, which is continuous and gradually declining in 3-D space, is employed to fill the pixel region ($\mathbb{R}^{8\times8}$), as calculated in (22a). With the generated Gaussian tensor centered on (x_i, y_i), the spatial continuum tensor is obtained by (22b) through a point-by-point combination. To account for the impact of distance from the concerned keypoint, the tensor is again globally optimized with the Gaussian function, resulting in the corresponding region pattern Ω_i, as described in (22c)

$$\mathfrak{G}(x, y; x_i, y_i, \sigma_1, \sigma_2) = \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{(x-x_i)^2+(y-y_i)^2}{2\sigma_1\sigma_2}} \quad (22a)$$

$$\Omega_i = \sum_{k}^{M} \mathfrak{G}\left(\left(p_i - p_i^k\right)_x + \frac{l}{2}, \left(p_i - p_i^k\right)_y + \frac{l}{2}, \rho, \rho \right) \quad (22b)$$

$$\Omega_i = \Omega_i \cdot \mathfrak{G}(l/2, l/2; \rho, \rho). \quad (22c)$$

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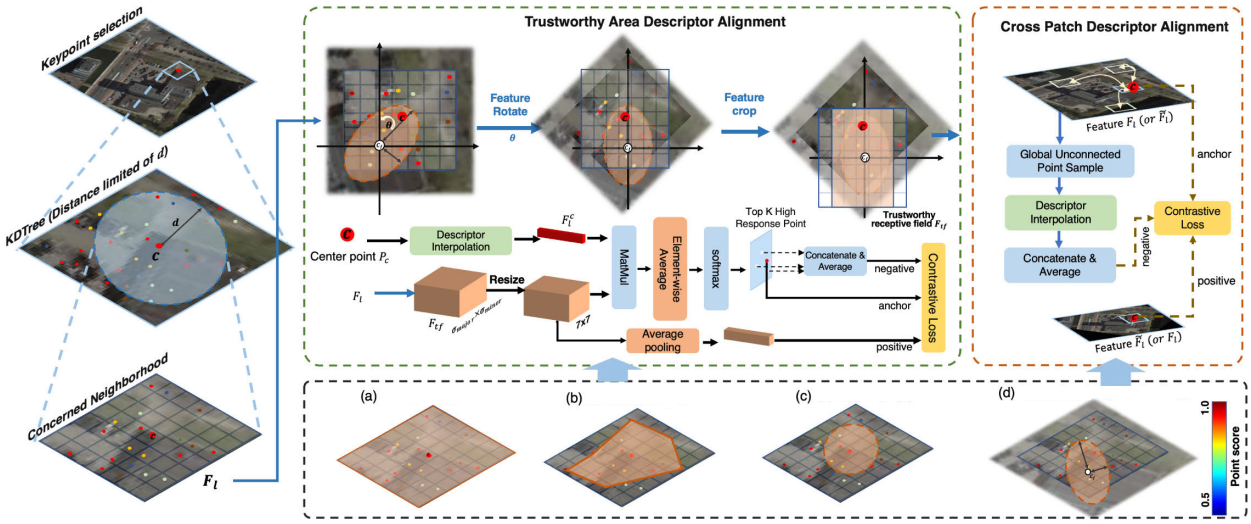


Fig. 7. Implementation of the descriptor alignment module that focuses on alignment within trustworthy areas and cross patches. (Left) Procedure of identifying keypoints and locating their neighbors with the KDTree algorithm.

Afterward, the region patterns are scaled to 8×8 and multiplied with the R^i_{fg}, preparing for loss calculation

$$W^i_p = \mathfrak{B}(\text{softmax}(\Omega_i);\ 8 * 8). \tag{23}$$

Emphasizing the correlation between morphological tensors and region patterns, pattern loss is designed to force the corresponding features of keypoint to align with relevant neighborhood patterns adaptively, as illustrated in the following:

$$L_{\text{KSF}} = \frac{1}{M} \sum_{p_i \in P_s \bigcup \widetilde{p}_i \in \widetilde{P}_s}^{M} \text{softmax}\left(R^i_{\text{fg}}\right) \cdot W^i_p. \tag{24}$$

The loss of neighbor space is defined as

$$L_{\text{Neighbor}} = L_{\text{CRFM}} + L_{\text{KSF}}. \tag{25}$$

E. Descriptor Alignment Module

Only focusing on the features of points themselves but overlooking the spatial entity where the keypoints exist is the unified limitation of the conventional detectors. Hence, expanding from neighbor windows, larger receptive fields about the existence entity and the whole images are concerned.

1) Existence Entity Construction: As shown in Fig. 7 (left), based on the neighbor relationship extracted from KDTree, the neighbor points of concerned point $p_i \in P_s$ are retrieved for entity construction under the distance limitation of $d = 28$. To eliminate unreliable interference, only keypoints with confidence above 0.5 are taken into consideration. As depicted in the bottom of Fig. 7, there are four solutions available for entity construction.

1) *Global Analysis:* Directly consider the 56×56 window and update the inner feature with the confidence of keypoints. However, redundant features are introduced.
2) *Convex Analysis:* Aggregate all neighbors with the convex hull function. The extreme point of the outermost convex hull has a great influence on region coverage.
3) *Center Analysis:* Delimit coverage with the multivariate Gauss $\mathfrak{G}$, and meanwhile, consider the keypoint confidence. Although suppressing distant keypoints, the useful features associated with the entity may be lost.
4) *Normalized Elliptic Analysis:* Rather than making keypoint the center of the belonging entity, a more reasonable approach is to reestimate the center with the neighborhood. With the aid of normalized elliptic analysis, the spatial morphology of the entity is characterized by the orientation and length of the major and minor semiaxes. It not only guarantees the robustness of the singularities but also maintains the inclusiveness of regional features.

The standard deviation ellipse is calculated as follows:

$$\bar{X}_i = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad \bar{Y}_i = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} \tag{26a}$$

$$\widetilde{x}_i^2 = \left(x_i - \bar{X}_i\right)^2 \quad \widetilde{y}_i^2 = \left(y_i - \bar{Y}_i\right)^2 \tag{26b}$$

$$\tan\alpha = \frac{\sqrt{\left(\sum_{i=1}^n w_i^2 \widetilde{x}_i^2 - \sum_1^n w_i^2 \widetilde{y}_i^2\right)^2 + 4\sum_{i=1}^n w_i^2 \widetilde{x}_i^2 \widetilde{y}_i^2}}{2\sum_{i=1}^n w_i^2 \widetilde{x}_i \widetilde{y}_i} + \frac{\left(\sum_{i=1}^n w_i^2 \widetilde{x}_i^2 - \sum_{i=1}^n w_i^2 \widetilde{y}_i^2\right)}{2\sum_{i=1}^n w_i^2 \widetilde{x}_i \widetilde{y}_i} \tag{26c}$$

$$\sigma_{\text{minor}} = \sqrt{\frac{\sum_{i=1}^n \left(w_i \widetilde{x}_i \cos\alpha - w_i \widetilde{y}_i \sin\alpha\right)^2}{\sum_{i=1}^n w_i^2}} \tag{26d}$$

$$\sigma_{\text{major}} = \sqrt{\frac{\sum_{i=1}^n \left(w_i \widetilde{x}_i \sin\alpha - w_i \widetilde{y}_i \cos\alpha\right)^2}{\sum_{i=1}^n w_i^2}} \tag{26e}$$

where the updated center is denoted as $(\bar{X}_w, \bar{Y}_w)$ and the orientation of the entity is determined by the angle α between the major axis and the y-axis in a clockwise direction. The lengths of the major and minor semiaxes are calculated with (26d) and (26e), respectively.

Considering $(\bar{X}_w, \bar{Y}_w)$ as the new center, we construct a minimum bounding rectangle based on the generated ellipse.

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This rectangle corresponds to the feature of the trustworthy field F_{tf}^i and is obtained by rotating F_l counterclockwise at an angle of α and then clipping it (mathematically denoted as $\mathfrak{T}$). The length and width of the rectangle correspond to the major axes σ_{major} and minor axes σ_{minor}, respectively,

$$F_{\text{tf}}^i = \Big\{ \mathfrak{T}(F_l; \alpha, \bar{X}_i, \bar{Y}_i) | \bar{X}_i - \sigma_{\text{minor}}^i < x < \bar{X}_i + \sigma_{\text{minor}}^i, \bar{Y}_i - \sigma_{\text{major}}^i < y < \bar{Y}_i + \sigma_{\text{major}}^i \Big\}. \quad (27)$$

2) Trustworthy Area Descriptor Alignment: In contrastive learning [53], the model optimization is accomplished by minimizing the distance loss about positive samples, negative samples, and anchors, as formulated in the following:

$$L_{\text{cons}}(f_{\text{anc}}, f_{\text{pos}}, f_{\text{neg}}) = \frac{1}{M} \sum_{p_i \in P_s \bigcup \widetilde{p}_i \in \widetilde{P}_s}^{M} \left[\alpha + \left\| f_{\text{anc}}^i - f_{\text{pos}}^i \right\|_2 - \left\| f_{\text{anc}}^i - f_{\text{neg}}^i \right\|_2 \right]. \quad (28)$$

This inspires us to convert the keypoint detection task into a pretext task that minimizes the feature difference between the extracted keypoint (anchor) and its existing domain (positive example). To eliminate unnecessary calculations on the entire image, the most disruptive vectors within the existing domain are defined as the hard negative examples.

Especially, for the sampled keypoint p_i, based on the credible entity features F_{tf}^i and the corresponding local vector f_l^i obtained from the local feature F_l, the following operations are conducted. First, we translate F_{tf}^i into regional describe features with the fixed size 7×7. Then, the positive samples f_{pos}^i are obtained through average pooling. For generating negative samples, we aim to find features that closely resemble the features of the observed window. Therefore, taking the inner product as the similarity measure, f_l^i is multiplied with each pixel on $F_{\text{region}}^i \in \mathbb{R}^{7\times7}$. Subsequently, elementwise average and softmax operations are performed. Excluding the originally detected keypoint, feature vectors of the top K similar pixels are selected as hard negative examples $\{f_{\text{neg}}^j\}_{j\in[1,K]}$. We summarize the generation of the credible region and the positive/negative samples into Algorithm 1. The descriptor alignment loss for features within the trustworthy entity area is defined in the following:

$$L_{\text{TrustArea}} = L_{\text{cons}} \left(f_l, f_{\text{pos}}, \frac{1}{K} \sum_{f_{\text{neg}}^j \in F_{\text{neg}}}^{K} f_{\text{neg}}^j \right). \quad (29)$$

3) Cross Patches' Descriptor Alignment: In order to analyze the correlation of features across the entire image, for each sampled point p_i, M uncorrelated keypoints are randomly selected as negative samples. Feature interpolation is performed to obtain descriptors of the negative samples $\text{NEG}^i = \{f_{\text{PatchNeg}}^j\}_{j\in[1,M]}$. Following the concatenation and channelwise averages, the negative samples are combined. As for the positive, under the guidance of repeatability, the feature descriptors $\widetilde{f}_l$ of the corresponding point $\widetilde{p}_i$ on the paired image are recognized. Based on these settings, we construct $L_{\text{CrossPatch}}$

$$L_{\text{CrossPatch}} = L_{\text{cons}}(f_l, \widetilde{f}_l, \mathcal{AP}_c(\text{NEG})). \quad (30)$$

Algorithm 1 Trustworthy Area and Sample Generation

Input: Extracted keypoints set P, local descriptor F_l
Output: F_{tf}: Region features of the induced entity, F_{neg}: Feature of the hard negative samples

$\mathcal{K} = KDTree(P)$;
Random Sample $P_s \subset P$;
foreach $p_i \in P_s$ **do**
  center descriptor: f_l^i;
  neighbor point set $P_{neighbor}^i = \{p_{ij} | p_{ij} \in P, \|p_i - p_{ij}\|_2 < d\}$;
  Calculate center $(\bar{X}_w, \bar{Y}_w)$ of $P_{neighbor}^i$, as Equ.(26a) ;
  Calculate orientation α, major axis σ_{major} and minor axis σ_{minor}, as Equ.(26c) (26d), (26e);
  Rotate $F_l : \mathfrak{T}(F_l; \alpha, \bar{X}_c, \bar{Y}_c)$;
  Select the trustworthy region F_{tf}^i, as Equ.(27);
  Generate region feature: $F_{region7*7}^i = \mathfrak{B}(F_{tf}^i, (7, 7))$;
  if *generate positive sample* **then**
    $f_{pos}^i = \mathcal{AP}(F_{region7*7}^i)$;
  else /* generate negtive sample */
    Obtain similar feature map: $F_{negmap} = softmax[\mathcal{AP}_c(f_l^i \cdot F_{region7*7}^i)]$;
    Select the top K position: $F_{neg}^i = topK\left(\max(\text{F}_{\text{negmap}})\right)$;
  end
end
Concatenation: $F_{tf} = \{\ldots, F_{tf}^i, \ldots\}$;
Concatenation: $F_{neg} = \{\ldots, F_{neg}^i, \ldots\}$;

Finally, the loss of the existence entity is defined as follows:

$$L_{\text{Entity}} = L_{\text{TrustArea}} + L_{\text{CrossPatch}}. \quad (31)$$

IV. Training and Inference

A. Training

Based on multigranularity spaces, three training rules are established. L_{Pixel} emphasizes the existence of keypoints (pixel window), L_{Neighbor} focuses on point distribution (neighbor window), and L_{Entity} uses contrastive learning to compare features within or across trustworthy domains (existence entity)

$$L = L_{\text{Pixel}} + \lambda_1 L_{\text{Neighbor}} + \lambda_2 L_{\text{Entity}} \quad (32)$$

where λ_1 and λ_2 are coefficients to balance cognition of points and descriptors. For clarity, the training procedure is illustrated in Algorithm 2.

In order to achieve efficient training, we make efforts in terms of data volume, relation generation, and computational techniques. Specifically, even though the multigranularity reason spaces and keypoint correlation are considered, only 30% keypoints are randomly sampled in the trustworthy keypoint induction module and the descriptor alignment module, optimizing computational efficiency without compromising performance. In addition, the precomputed KDTree reduces redundant computations, acting as uniform inputs for several

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Algorithm 2 Training Procedure

Input: Images $\{I_0, I_1, \ldots\}$, random matrixes $\{H_0, H_1, \ldots\}$
Output: Network $Model(\cdot)$

Initialize $Model(\cdot)$ with ImageNet-pre-trained VGG16;
Initialize the decoded network ;
for *epoch = 1* **to** *MaxEpochs* **do**
  for *i = 1* **to** *MaxIters* **do**
    Convert I_i to $\widetilde{I}_i$ with H_i;
    Obtain S and W_s; Obtain $\widetilde{S}$ and $\widetilde{W_s}$;
    Obtain P and $\widetilde{P}$ as Equ. (8), (9), (10);
    Co-visible points Screen as Equ.(12);
    Extract F_{fg}, F_l;
    $\mathcal{K} = KDTree(P)$,$\widetilde{\mathcal{K}} = KDTree\left(\widetilde{P}\right)$;
    P_s sampled from U,$\widetilde{P}_s$ sampled from V, with ratio of 30%;
    for $p_i \in P \bigcup \widetilde{P}$ **do**
      if *in pixel window* **then**
        Calculate L_{Ref} as Equ. (13b);
        Calculate $L_{RegionRel}$ as Equ. (14);
        Calculate L_{Heter} as Equ.(18);
        Loss summarizes: Equ. (19);
      if *in neighbor windows and* $p_i \in P_s \bigcup \widetilde{P}_s$ **then**
        Do DFSS. Obtain X_c, X_n, and X_d, as Equ. (20d);
        Calculate L_{CRFM} as Equ. (21);
        Generate region pattern as Equ. (22);
        Calculate L_{KSF} as Equ. (24) ;
        Loss summarizes: Equ. (25);
      if *in existence entity and* $p_i \in P_s \bigcup \widetilde{P}_s$ **then**
        Conduct algorithm **1**;
        Calculate $L_{TrustArea}$ and $L_{CrossPatch}$;
        Loss summarizes: Equ. (31);
      Loss merge: Equ.(32);
    end
    Update $Model(\cdot)$ with the loss;
  end
end

modules. Furthermore, we leverage parallel computing techniques and GPU acceleration, effectively utilizing available computational resources during training. Notably, modules other than the codec network and subsequent coarse keypoint regression head are not applied during inference, ensuring optimal efficiency in keypoint detection.

B. Inference

To eliminate features without scene semantics, regional reliability descriptor W_s is used to update the score map

$$S_w = \mathbf{1}[\mathfrak{B}(W_s, (W, H)) > \theta)] \odot S \tag{33}$$

where $\mathbf{1}$ is the matrix filled with ones, W and H are the size of the original image, and $\odot$ is the Hadamard product. The value is set to zero when W_s is less than threshold θ, meaning regions without explicit semantics. During inference, the top K positions with the highest response in the optimized score are selected as the keypoints (1500 in our work).

Algorithm 3 Inference Procedure

Input: Image I, network model $Model(\cdot)$
Output: Keypoint set P

Load training model $Model(\cdot)$;
Input image I;
Obtain score map S, regional reliability tensor W_s;
Update score map with Equ. (33);
Get the position of keypoints as Equ. (8), (9), (10);

V. Experiment

A. Datasets

In order to evaluate TSK and other keypoint detectors, one self-collecting dataset and three change detection benchmarks of RSI are employed.

1) *Multitemporal Images Collected From Google Earth (MICGE) [3]:* The dataset consists of 13 groups of RSIs, in which shooting areas include rivers, towns, wharves, and so on. Resolutions of images are about 1 m, and sizes range from 6000 to 10 000 pixels.
2) *Onera Satellite Change Detection (OSCD) Dataset [26]:* 24 multispectral images taken from the Sentinel-2 satellite in 2015 and 2018 are included. The spatial resolution varies between 10, 20, and 60 m. The performance of this dataset provides insights into the detector's capability to handle mid- and low-resolution RSIs.
3) *High-Resolution Semantic Change Detection (HRSCD) Dataset [27]:* This dataset contains 291 coregistered image pairs of RGB aerial images from IGS's BD ORTHO database. It describes changes in city-wide land use of the European Union in 2006 and 2012. It assesses the effectiveness of the keypoint detectors for multitemporal images exhibiting significant changes.
4) *SZTAKI Dataset [28]:* The dataset contains 13 pairs of aerial images with a size of 952 × 640 pixels and a spatial resolution of 1.5 m. The dataset encompasses various changes, including the emergence of new urban areas, extensive tree planting, changes in cultivated land, and reconstruction.

B. Metrics

Inheriting from the standard registration pipeline of "keypoint detection–descriptor extraction–descriptor matching–matrix solving," results of matching and matrix solving are used to evaluate the trustworthiness of keypoints generated by the TSK detector. Reiterating the definition of CKpts as the previous works [9], [22], after coordinating transformation with the predicted matrix H, if the distance of a pair of keypoints is less than threshold t, the specified keypoints are considered as CKpts.

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1) *Repeatability Score (RS):* The ratio of the amount of CKpts to the amount of all detected keypoints is defined as RS. Only Ckpts presented in the common-observation area participate in the calculation. The final results are obtained by taking an average of all views and all point pairs.
2) *Localization Error (LE):* For Ckpts, whose distance is less than threshold t, the root mean square error (RMSE) is utilized to measure the location offset of per point pair. Just like RS, the average of all views and all pairs is defined as the location error.
3) *Keypoint Matching Rate (KMR):* Detecting more useable Ckpts for the subsequent matching is the goal of keypoint detection in the registration pipeline. To measure the contribution of the keypoint detector to matching, KMR is designed, which is defined as the ratio between the amount of Ckpts identified by the matching algorithm to the amount of all CKpts limited by distance threshold t.
4) *RMSE of the Matched Points (MRMSE):* Evaluating from the perspective of matching, the mean RMSE among the matched keypoints is calculated.
5) *Mean Matching Accuracy (MMA):* The ratio of the amount of the matching key pairs, whose RMSE is less than the threshold t, to the amount of all matched pairs.
6) *Homography Error (HE):* The average distance between the angular points of original images and registered images is used to evaluate the accuracy of the solved transformation matrix H.

C. Implementation Details

Our training set is created by mixing parts of the natural image dataset "revisitop1m" [54] with the RSIs in a ratio of 8:2. It provides a broader representation of visual features, allowing the model to better handle variations in both natural and RSIs. We crop the training images into patches with size 512×512 and produce the corresponding auxiliary images with the randomly generated rotation matrix H, simulating multitemporal images captured from different camera views. It is assumed that the matrix satisfies the following randoms conditions: rotation angle ranges from $-45°$ to $45°$, shrinkage scale ranges from 0.5 to 2.0, and the coefficient of skew ranges from -0.8 to 0.8. In addition, color migration, salt and pepper noise, and color channel changes are applied to drive diverse image representations.

We implemented our TSK method in PyTorch with the graphics computing acceleration device GTX 2080 Ti. The feature extraction and fusion network consists of hierarchical blocks F^i_{ms} with channel numbers 64, 128, 256, and 512, respectively. These channel numbers maintain symmetrical consistency with the deconvolution process. The generated fine-grained descriptors F_{fg} and local descriptor F_l both have dimensions of 128. The kernel size of the convolution kernel is 3, and the activation function is ELU, with the hyperparameter $\alpha = 1$. During training, we set the SGD optimizer with a fixed learning rate of 5e-2, a momentum of 0.5, and a weight decay of 1e-5. The training consists of 5 epochs, and the batch size for both training and evaluation is consistently set to 1. During inference, we set $k = 1500$ as the max amount of keypoints, the confidence threshold θ of the regional reliability tensor W_s is set as 0.3, and the loss balance parameters $\lambda_1 = \lambda_2 = 1$.

D. Qualitative Evaluation

We present comparative evaluations and diagnostic experiments on the mentioned datasets with our TSK and other state-of-the-art detectors. The detection results are summarized in Fig. 8. Each group of images is divided into upper and lower parts. The above part shows the distribution of the detected keypoints in two images, where white indicates the absence of the corresponding point, while other colors represent their presence. The bottom part represents the matching process of keypoints. Colored points connected with lines are matched keypoints, and their color follows the jet color bar according to their confidence predicted from score map S. The white points not connected are the points without the matchings. Note that our experiments are conducted under the threshold of $t = 3$.

1) Effect of Keypoint Extraction: The top part of Fig. 8 gives a clear view to analyze the location and density of the detected keypoints. For the handcrafted detectors FAST and SIFT, most corners, ridges, and boundaries that exhibit intense gradient changes are recognized. However, keypoints on the boundaries are not evenly distributed, which leads to a large number of points that cannot be judged as repeatable. At the same time, the texture with high gradients intense on forest, farmland, and lawn results in several invalid keypoints. For D2-Net, R2D2, and LoFTR, which distribute keypoints throughout the entire image due to the grid arrangement, a large number of unexplained keypoints are located outer black edge or on the image boundary and areas with weak texture. It is evident in the results of R2D2 and LoFTR, such as the uniform distribution keypoints in the river channel (case a) and farmland (case d). In contrast, the sparse keypoints generated by SuperPoint and Key.Net seem more reasonable and semantical; however, the presence of keypoints on the image edge, grass, and river channel indicates that some of their detection results are still meaningless. It is speculated that the recognition of keypoints is limited by the incomplete handcrafted annotation. Compared to other options, the keypoints generated by the TSK detector are the most reasonable. The keypoints are not only distributed on the edges and corners of houses (case a) but also within the salient targets (case c). As for edges, TSK attempts to generate dense results (approximately a complete line) for higher repeatability. Meanwhile, as a learning-based scheme, unlike D2Net, R2D2, and SuperPoint, no keypoint is detected on the image boundary in TSK results, demonstrating a clear understanding of real semantic points rather than interference.

2) Reliability for Keypoint Matching: Matching results are shown at the bottom of each set of images of Fig. 8. Note that, for the dedicated keypoint detectors that focus on keypoint detection but ignore matching procedures, such as FAST, SIFT, Key.Net, and TSK, we make HardNet [55] the common matching descriptors extractor and mutual nearest neighbor (MNN)

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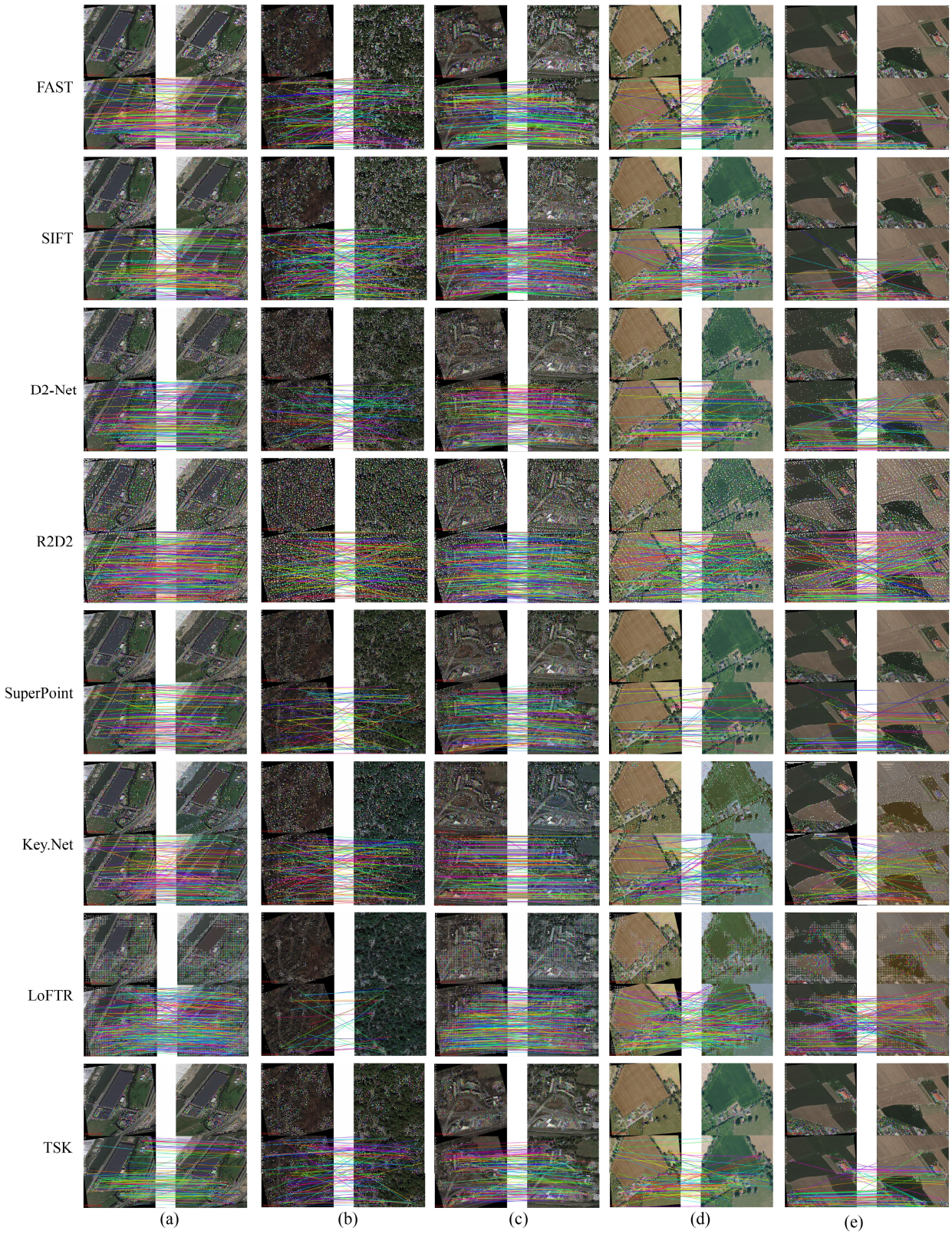


Fig. 8. Keypoint detection results of different methods. Each set of images contains four subgraphs. The above two demonstrate the distribution of the detection keypoints in the multitemporal images, where the white indicates the absence of corresponding points in another image, while colored points represent the presence of corresponding points. The bottom two images represent the matching process. Colored points connected by lines are matched keypoints, and their color follows the Jet color bar according to their confidence predicted from score map S. (a)–(e) Five representative scenes.

the descriptor matcher. For SuperPoint, D2-Net, and R2D2, their descriptors are extracted synchronously during keypoint generation, serving as the matching descriptors as the original working requirements. Generally speaking, regions with weak semantics (forests, farmlands, and lawns) are merely matched. Even if they are matched, there is a high probability of errors.

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TABLE I
PERFORMANCE COMPARISON FOR KEYPOINT DETECTORS

Method	RS ↑	LE ↓	KMR ↑	MRMSE ↓	MMA ↑	HE ↓	RS ↑	LE ↓	KMR ↑	MRMSE ↓	MMA ↑	HE ↓
	MICGE dataset						HRSCD dataset					
FAST†[12]	0.378	1.314	0.726	0.611	0.980	1.385	0.147	1.841	0.378	**1.453**	0.311	1.367
SIFT†[30]	0.399	1.357	0.627	0.669	0.978	1.305	0.237	1.892	0.277	1.700	0.360	2.353
SuperPoint[22]	0.405	1.440	0.858	0.835	0.968	1.806	0.095	1.906	**0.527**	1.577	0.219	1.573
D2-Net[8]	0.209	1.916	0.666	1.529	0.797	3.923	0.149	1.960	0.248	1.714	0.231	1.397
R2D2[9]	**0.619**	1.390	0.328	1.908	0.977	1.673	0.185	1.922	0.101	1.693	0.183	1.448
Key.Net[21]	0.505	1.291	0.466	0.702	0.989	1.141	0.168	1.840	0.207	8.138	0.246	1.689
LoFTR[47]	0.333	1.929	0.657	1.238	0.891	3.343	**0.334**	1.919	0.340	5.609	0.217	1.503
TSK†	0.494	**1.271**	**0.867**	**0.601**	**0.991**	**1.222**	0.295	**1.776**	0.398	1.571	**0.372**	**1.302**
	OSCD dataset						SZTAKI dataset					
FAST†[12]	0.513	0.980	0.837	0.213	0.986	0.495	0.381	1.597	0.491	0.980	0.846	2.285
SIFT†[30]	0.468	0.998	0.655	0.224	0.985	0.528	0.345	1.712	0.384	1.064	0.832	1.851
SuperPoint[22]	0.518	1.178	0.849	0.302	0.985	0.861	0.232	1.704	0.535	1.149	0.772	2.456
D2-Net[8]	0.200	1.878	0.687	1.369	0.847	4.092	0.174	1.927	0.349	1.513	0.599	3.637
R2D2[9]	**0.625**	1.180	0.352	0.419	0.986	0.714	**0.441**	1.561	0.147	1.284	0.703	1.949
Key.Net[21]	0.574	0.944	0.497	0.343	0.984	0.527	0.342	**1.473**	0.326	**0.896**	**0.901**	1.801
LoFTR[47]	0.272	1.968	0.741	9.246	0.750	2.753	0.376	1.835	0.410	1.285	0.798	3.064
TSK†	0.528	**0.892**	**0.916**	**0.172**	**0.987**	**0.484**	0.362	1.537	**0.637**	0.917	0.871	**1.512**

* Bold text denotes the best performance. The upward arrow signifies that a higher value indicates better performance, while the downward arrow indicates the opposite.
** Mutual nearest neighbor (MNN) is used to match descriptors, and RANSAC is used for all comparative methods for wrong matching inhibition.
† HardNet [55] is employed as the descipritor extractor.

With the increase in false matches, the following transformation matrix solving will be seriously disturbed, which is evident in the results of SIFT and R2D2. Notably, LoFTR is an algorithm prioritizing matching; its performance is not ideal in RSIs characterized by low texture and densely distributed targets. In addition, due to unclear point recognition, a significant number of unnecessary matching operations contribute to time inefficiencies. Comparing TSK with SuperPoint and Key.Net, under the consistent matching quantity, the keypoint density of TSK is significantly low. The keypoints detected by TSK are few but fully used, which provides additional evidence that the TSK-related keypoints are semantically sufficient and trustworthy.

3) Reliability for Registration: After applying RANSAC for matching outlier suppression, the transformation matrix is solved. The checkerboard results of different keypoint detectors are summarized in Fig. 9. The most noteworthy areas are marked with red boxes. Owing to the unfavorable factor of huge color changes and repetitive textures, D2-Net completely fails in the registration of case (a), and FAST and R2D2 are not perfectly aligned. For cases (b) and (c) with the distribution of multitype targets (i.e., buildings and forests), SIFT and R2D2 showed obvious registration errors. Against expectations, each method performs better under the road scene, but subtle registration errors are also observed in SuperPoint and D2-Net. Regardless of the circumstances, the registration based on the keypoints extracted from TSK is consistently outstanding.

E. Quantitative Evaluation

The performance for keypoint reliability (RS and LE), matching validity (KMR, MRMSE, and MMA), and registration accuracy (HE) is collected in Table I. It is worth noting that the TSK detector outperforms all other keypoint detectors from the comprehensive performance.

1) Effectiveness of Keypoint Extraction: RS and LE are arranged in the first two columns of Table I. TSK detector demonstrates universal adaptability for all RMI datasets, particularly the HRSCD dataset with complex target distribution in a large viewing field. In the MICGE dataset, despite the highest RS is not acquired, TSK outperforms the other methods in LE (two-thirds of the worst D2-Net). As expected, R2D2 achieved remarkable RS across all datasets, especially in OSCD and SZTAKI. As shown in Fig. 8, R2D2 and LoFTR ensure points placed in every search window, which increases the probability of random points finding their correspondings satisfying the threshold $t = 3$. This leads to a high RS. When interpreted together with LE, R2D2 and LoFTR fall to the second-to-last and third-to-last ranking, but the TSK detector maintains the first place.

2) Reliability for Keypoint Matching: Rather than quantity, the quality of keypoint is more prominent for the subsequent matching and registration. Therefore, KMR, MRMSE, and MMA are applied. As shown in Table I, TSK maintains the top two ranks in KMR among all datasets. It is proven that the keypoints generated by the TSK detector are distinguishable and semantically rich. Furthermore, as discussed before, R2D2 and LoFTR demonstrate high RS but significantly low KMR. This suggests that the majority keypoints of the two methods serve as meaningless interference points for matching. Comparatively, the TSK detector generally exhibits superior KMR to others. Specifically, TSK achieves a KMR of 0.916 for the OSCD dataset, which is 2.6 times higher than that of R2D2. Notwithstanding multitemporal images with various categories

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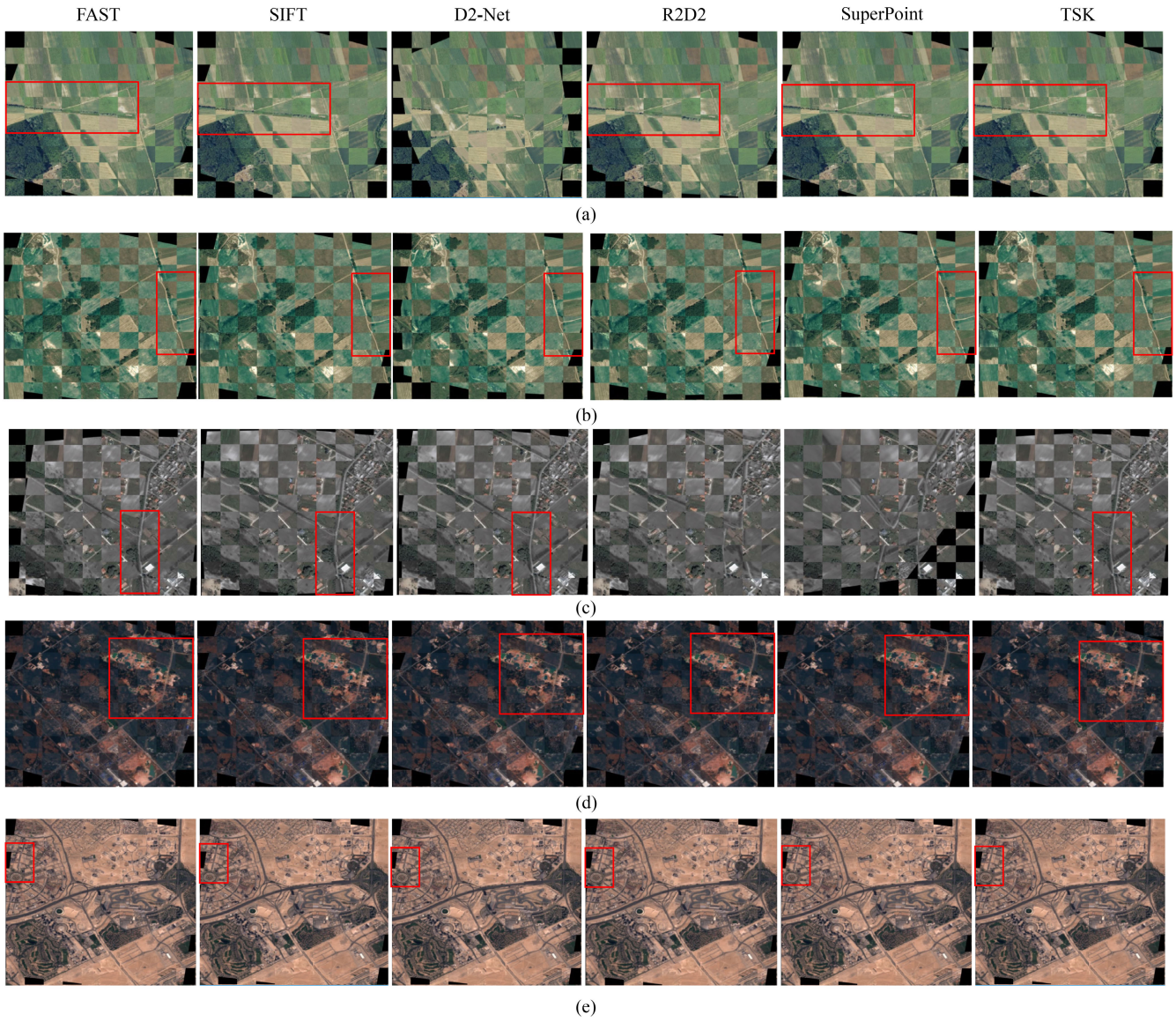


Fig. 9. Registration results based on the keypoints detected from different detectors. The red boxes highlight the significant areas, and the misaligned textures indicate registration failures. (a)–(e) Five representative scenes.

(MICGE dataset) or large-scale variation (OSCD dataset), TSK is almost the best in MRMSE and MMA. While the optimum performance is not achieved in HRSCD, the results of all other methods hover around 1.6 with low variance.

In addition, compared to FAST, which performed the best in MRMSE, TSK showed a significant improvement in MMA from 0.311 to 0.372. It is inferred that the slightly low MRMSE is negligible for questioning the robustness of TSK. In terms of SZTAKI, Key.Net demonstrates outstanding performance with LE, MRMSE, and MMA (TSK maintains the second). However, in the final registration, TSK exhibits a relative advantage in HE over Key.Net, which underscores TSK's robustness and beneficial contribution to subsequent tasks. It is worth emphasizing that while D2-Net, SuperPoint, and LoFTR construct descriptors by integrating keypoint detection with descriptor matching, TSK solely relies on the baseline HardNet for descriptor extraction. It provides evidence that the TSK relaxes the capacity requirements of matching modules, and the high-precision matching result is capital to be achieved by few trusted keypoints together with the basic matching algorithms.

3) Reliability for Registration: HE measures the difference between the solved matrixes and the annotations. It can be observed that the value distribution of HE exhibits high variance across all methods and datasets, indicating that the choice of keypoint detector has a significant impact on the registration process. TSK is the most outstanding method that controls the HE errors within 1.6. It even achieves subpixel-level registration in OSCD, providing evidence that the keypoints are semantically effective for registration.

4) Model Efficiency: We conducted experimental analysis on floating point operations per second (FLOPS), model parameter quantity, and inference time, as collected in Table II. The 512×512 images are applied as inputs, and the

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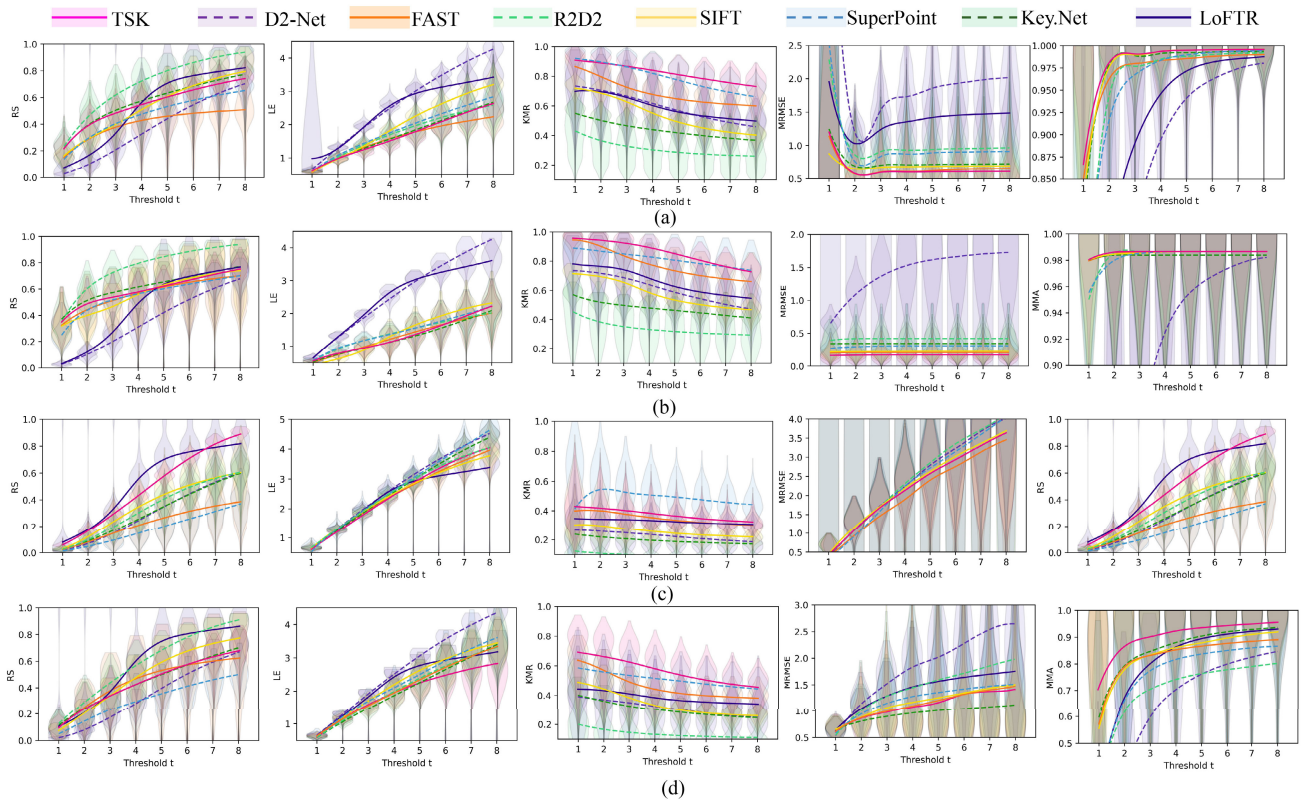


Fig. 10. Performance comparison diagram of the competitive keypoint extraction methods and TSK detector under different evaluation threshold t. Each curve is fit with the average values of all results under a specific threshold t, and the color-filled area represents the frequency histogram, derived from the statistical analysis of the performance for each pair of images. Since the relaxed threshold makes no significant numerical variation on HE, the HE-related curves are omitted. (a) MICGE dataset. (b) OSCD dataset. (c) HRSCD dataset. (d) SZTAKI dataset.

TABLE II
EFFICIENCY OF THE LEARNING-BASED KEYPOINT DETECTORS

Method	FLOPs (G)	Parameters (M)	FPS	TPIP (s)
SuperPoint[22]	20.841	1.300	1.324	0.755
D2-Net[8]	134.867	7.635	8.265	0.121
R2D2[9]	119.183	0.484	5.870	0.170
Key.Net [21]	3.174	0.006	2.166	0.462
LoFTR [47]	320.954	11.561	1.425	0.702
TSK	191.887	12.797	4.290	0.233

computing device is a GTX 2080 Ti graphics computing acceleration device. Among the detectors, Key.Net achieves the lowest FLOPS due to its operation on single-channel gray-scale images and shallow feature extraction networks. D2-Net and R2D2 have slightly higher FLOPS and model parameters. High FLOPS and model parameters are exhibited in LoFTR due to the incorporation of transformer-related global attention mechanisms. In comparison, despite the utilization of multiple modules and cross-layer connections, the model parameters and FLOPs of TSK are acceptable, making it feasible for operation in most application environments. Following the pipeline of "keypoint detection-matching-matrix solving," frames per second (FPS), and time per image pair (TPIP) are used to measure the registration time. The excessive matching and complex computations contribute to suboptimal time efficiency in LoFTR. However, TSK maintains an acceptable computational speed, even surpassing the Key.Net with fewer parameters. It performs comparably to the R2D2 method and is 3.2 times faster than the SuperPoint. These results confirm that TSK does not introduce significant delays in keypoint detection, even when it performs feature refinement on multiple spatial granularities. Overall, TSK strikes a balance between computational efficiency and accuracy, making it a viable solution for various real-world scenarios.

F. Parameter Analysis

1) Evaluation Threshold: To eliminate the one-sidedness of performance evaluation caused by the fixed threshold, we take t as the variable to further demonstrate the potency of TSK. The performance comparison diagrams of various datasets are shown in Fig. 10. Each curve is fit with the average values of all results under a specific threshold t, and the color-filled area represents the frequency histogram, derived from the statistical analysis of the performance for each image pair. Since the relaxed threshold makes no significant numerical variation on HE, the HE-related curves are omitted.

1) *RS:* TSK is prominent among all datasets, especially for the HRSCD dataset. In fact, when the threshold is relaxed to 8, a noticeable upward trend appears in R2D2 and LoFTR, which verified our speculation that excessively dense keypoints make RS lose fairness.
2) *LE:* Contrary to the distribution observed in RS, R2D2 and LoFTR exhibit poor performance in LE. However, the TSK detector maintains superb performances in RS

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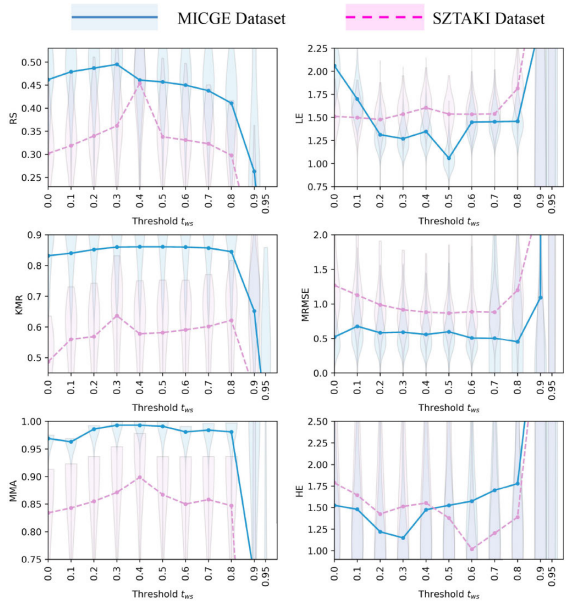


Fig. 11. Trend graphs of all performance indicators for adopting different thresholds t_{ws} on W_s.

and LE. Especially in SZTAKI and OSCD, there is no greater change even if the threshold is relaxed. This finding supports the notion that the keypoints generated by TSK have a more definitive position induction, rather than being randomly generated based on texture strength.

3) *KMR:* For HRSCD, the performance of TSK is preferable and shows a small gap with the optimal SuperPoint. In other datasets, TSK has surpassed Superpoint. However, R2D2 yields consistently low and nearly zero results.
4) *MRMSE:* MRMSE is an updated result of LE obtained by screening the most probable matching descriptors. Due to the absence of fine-tuning based on the specific dataset, LoFTR encounters numerous mismatches, resulting in excessively high MRMSE. It is found the MRMSE of TSK is always low whatever the datasets or threshold, which demonstrates its high adaptability across diverse datasets.
5) *MMA:* TSK outperforms most methods in all datasets. It is attributed to the comprehensive learning of the multiscale and full-space features. In contrast, D2-Net and R2D2 only analyze the keypoint feature within the observed window, without recognizing the environmental feature of other areas.

2) Threshold of W_s: The regional reliability tensor W_s is used to update the score map S during inference. To investigate the reasonable threshold, MICGE and SZTAKI are chosen as representative datasets for evaluation. Results are illustrated in Fig. 11. The incremental threshold setting is performed at intervals of 0.1 starting from 0. Since no keypoint remains when $t_{\mathrm{ws}} = 1$, the last statistic was conducted at $t_{\mathrm{ws}} = 0.95$. RS reaches the peak when $t_{\mathrm{ws}} \in [0.3, 0.5]$ and then slightly decreases. Since the confidence of W_s is concentrated in 0.5–0.8, the decline is not significant. A sharp decline occurs when t_{ws} is greater than 0.8, which can be attributed to the feature suppression of the majority keypoints. Similar trends are also observed in KMR and MMA. In contrast, LE, MRMSE, and HE exhibit a fluctuating decline followed by a sudden rise. Typically, the optimal results are found within the range of 0.2–0.6. Hence, it is believed that $t_{\mathrm{ws}} \in [0.2, 0.5]$ is suitable. We consider setting $t_{\mathrm{ws}} = 0.3$ as a reasonable choice.

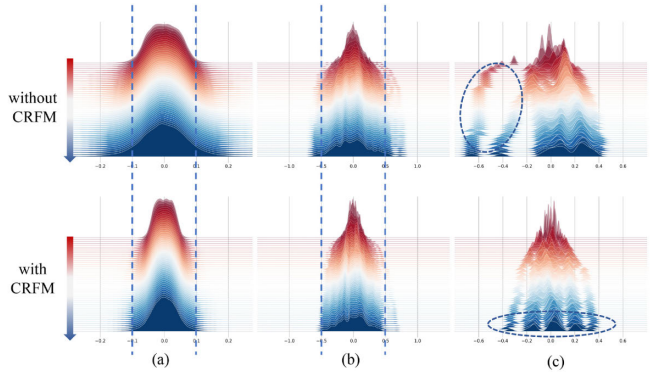


Fig. 12. Ridgeline diagrams of the convolution kernel. Each diagram represents the changes in parameters during training from (top) to (bottom). The upper row originates from the model trained without CRFM, and the lower row originates from the model with CRFM. (a) Histogram of weights for the deconvolution kernel associated with F_d^1. (b) Histogram of weight for the deformable convolution kernel. (c) Histogram of bias for the deformable convolution kernel.

G. Ablation Study

1) Effects of CRFM: As collected in Table III, little improvement or even a reduction of RS shows when CRFM is the only supplementary component. However, MRMSE and HE exhibit upward trends instead. It is hinted that although CRFM provides a larger receptive field and stronger local representation, it is susceptible to confusion for keypoints recognition. To illustrate how the replication process facilitates the response to multifrequency features, various histograms about the weight and bias of convolution kernels are compared.

1) *Deconvolution Kernel Associated With F_d^1:* As shown in Fig. 12(a), the kernel weight with CRFM shows centrality to zero compared to that without CRFM. The sparsity of convolution kernel parameters is improved. It can be inferred that CRFM helps the network suppress the interference on the first step of point feature refinement.
2) *Deformable Convolution Kernel:* As the crucial step in keypoint localization, we visualize the weight of the deformable convolution kernel in Fig. 12(b). After applying CRFM, the weight of the deformable convolution kernel becomes more concentrated around 0, indicating that more sparse features are produced along with better interference suppression. An even starker difference is seen in bias, as shown in Fig. 12(c). Before CRFM is used, most kernels with bias aggregate around $[-0.1, 0.35]$, and only a small part concerns the low potential. It is inferred that the high-frequency texture is emphasized. After applying CRFM, the bias distribution becomes more symmetric and dispersed, and feature responses with multilevel potential are output. It can

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TABLE III
PERFORMANCE COMPARISON TABLE: ABLATION EXPERIMENTS ON TSKS TRAINED WITH DIFFERENT COMBINATIONS OF COMPONENTS

			RS ↑	LE ↓	KMR ↑	MRMSE ↓	MMA ↑	HE ↓	RS ↑	LE ↓	KMR ↑	MRMSE ↓	MMA ↑	HE ↓
CRFM	**KSF**	**DA**	MICGE dataset						HRSCD dataset					
×	×	×	0.490	1.348	0.837	0.630	0.989	1.242	0.208	1.918	0.395	1.696	0.308	1.997
√	×	×	0.492	1.307	0.840	0.636	0.987	1.247	0.208	1.918	0.397	1.683	0.365	1.957
×	√	×	0.472	1.281	0.848	0.600	0.987	1.295	0.188	1.916	0.404	1.633	0.327	1.745
×	×	√	0.491	1.319	0.837	0.633	0.988	1.232	0.212	1.916	0.397	1.645	0.368	2.156
√	√	×	0.470	1.284	0.847	0.602	0.988	1.307	0.185	1.917	0.408	1.628	0.329	1.926
√	×	√	0.475	1.322	0.849	**0.595**	0.986	1.247	0.177	1.912	**0.409**	1.650	0.370	2.143
×	√	√	0.474	1.329	0.848	0.614	0.984	1.258	0.211	1.909	0.407	1.648	0.329	1.955
√	√	√	**0.494**	**1.271**	**0.867**	0.601	**0.991**	**1.222**	**0.295**	**1.776**	0.398	**1.571**	**0.372**	**1.302**
CRFM	**KSF**	**DA**	OSCD dataset						SZTAKI dataset					
×	×	×	0.524	1.000	0.902	0.233	0.968	0.624	0.360	1.589	0.617	1.068	0.841	1.963
√	×	×	0.523	1.002	0.915	0.215	0.977	0.602	0.358	1.584	0.620	1.030	0.865	1.982
×	√	×	0.550	0.894	0.934	0.196	0.982	0.493	0.321	**1.540**	0.632	0.932	0.807	1.736
×	×	√	0.526	1.002	0.916	0.212	0.972	0.668	0.322	1.583	0.617	1.037	0.805	1.821
√	√	×	**0.553**	0.905	0.934	0.197	0.987	0.518	0.326	1.552	0.630	0.955	0.862	1.709
√	×	√	0.552	**0.893**	**0.935**	0.193	0.980	0.618	0.321	1.586	0.617	1.058	0.791	1.676
×	√	√	0.553	0.894	0.934	0.194	0.982	0.507	**0.364**	1.538	0.634	0.960	0.807	1.690
√	√	√	0.528	0.892	0.916	**0.172**	**0.987**	**0.484**	0.362	1.537	**0.637**	**0.917**	**0.871**	**1.512**

* Bold text denotes the best performance and underscore denotes the worst.

be inferred that CRFM allows the subsequent decoder networks to integrate multifrequency information, providing a sparse and uniform feature performance for subsequent the keypoint location.

2) Effects of KSF: KSF refines the local fine-grained feature with both location and confidence of neighbor keypoints. Taking RS and LE as examples, as shown in Table III, the experiments on the SZTAKI dataset confirm that the KSF module does enhance the localization effect (the optimal LE). It indicates that the neighbor keypoints are useful for strengthening features of the analyzed central keypoint. However, it still requires synergy with other modules to prevent being confined to point topology recognition and overlooking local semantics. Besides, improvements are also revealed in the matching and transformation matrix solving. Unexpectedly, a worse effect is obtained when KSF combines with CRFM or DA components alone. It is inferred that solely combining CRFM or DA is inadequate for constraining the features, resulting in the confusion of local semantics and the generation of unreliable neighborhood keypoints.

3) Effects of DA: The most notable enhancement to the network arises from this module, as it simultaneously takes into account representativeness and the distinction between features with or without explicit semantics. However, it can indeed lead to a potential decrease in localization accuracy when used in isolation. However, as shown in Table III, even when used alone, the performances are close to the case of using all components. Especially with the help of CRFM, the multiscale and multifrequency features are comprehensively considered, which output prominent capability in keypoint detection in OSCD datasets. Building upon this, by paying attention to the position and features of neighboring keypoints (KSF), there has been a significant improvement in RS and LE. In the OSCD dataset, the RS improved from 0.523 to 0.553 and SZRAKI from 0.321 to 0.364. Therefore, we conclude that refining features of keypoints from multispatial granularity is effective.

H. Feature Visualization for Trustworthy Validation

In order to facilitate a comprehensive understanding of TSK's functionality and advance it to a more transparent and interpretable solution, some essential features are visualized in Fig. 13. To enhance our understanding of these features, we employed feature gradient maps as a supportive tool. Specifically, the local descriptor F_l, the regional reliability weight W_s, the first decoder feature F_d^1, the penultimate decoder feature F_d^3, the last decoder feature F_d^4, and the generated score map S are analyzed. An ablation study is conducted to evaluate the impact of the CRFM, KSF, and DA.

1) Discussion on Local Descriptor: The encoded local descriptor F_l serves as the foundation for subsequent calculations, playing a crucial role in determining the positional accuracy and semantic validity of keypoints. It is imperative for F_l to encapsulate as much as possible information. As shown in Fig. 13, the local descriptor F_l of the baseline TSK exhibits a relatively uniform distribution. According to information entropy principles, more evenly distributed information signifies richer content contained. Shifting our focus to the corresponding gradient image, the evidence edge, block, and even the inconspicuous texture can be observed. For the model trained with only KSF, conspicuous responses are observed as dotted lines on the road and discrete houses. Despite utilizing neighboring features, the limited receptive fields result in a preference for high-frequency textures. To facilitate comparisons, we employed a combination of DA and CRFM. Specifically, its F_l exhibits improved uniformity; meanwhile, its gradient map contains more explicit scene perception. Especially, for the grass area, the absence or weak presence of gradient proves the trustworthy semantic in keypoint detection. Furthermore, the local descriptor of the complete version demonstrates better uniformity. The enhanced response of detailed features in gradient maps further validates the rationale for refining key features through multigranularity windows.

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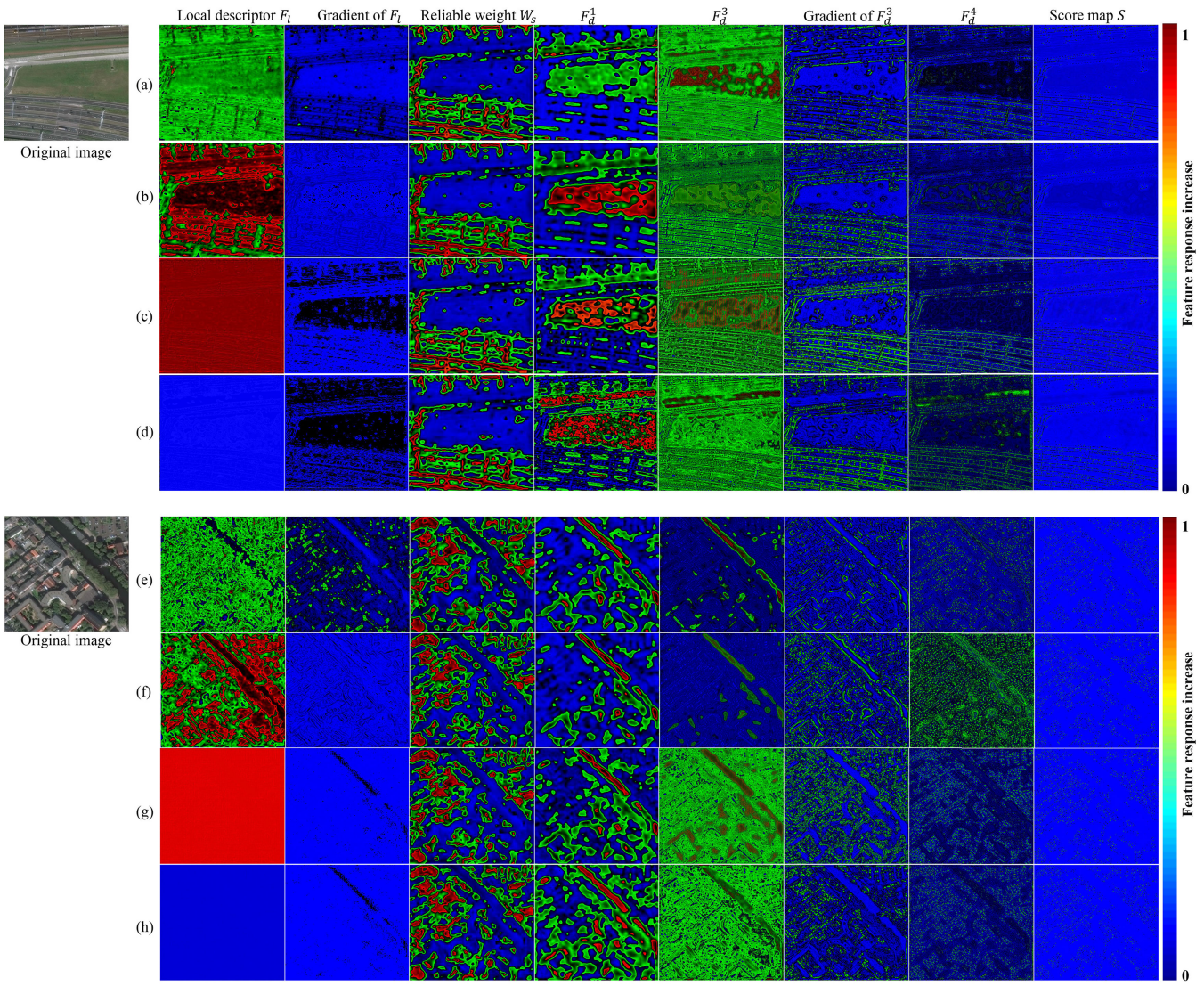


Fig. 13. Feature visualization of different stages trained with different component combinations. (a) and (e) Baseline TSK, training without any CRFM, KSF, or DA component. (b) and (f) Training only with KSF. (c) and (g) Training with CRFM and DA. (d) and (h) Training with all components.

2) Discussion on Decoding Features: For the first decoding layer F_d^1, more refined features are presented, thanks to the synergistic action of the three components. The degree of feature depiction has been enhanced, which is intuitively demonstrated in the transition from emphasizing the overall road edges to being capable of clearly identifying each road sign. Interestingly, there are also noticeable high intensities present in the lawns of case (a). Given the usage of skip connections in the network architecture, we cautiously deem such a distribution to be reasonable.

As the model evolves, significant improvements are also observed in F_d^3. The incorporation of components introduces spatial attention with multiple spatial granularities, compelling the network to update from the original sporadic keypoint response to the subsequent strong response. Analyzing the gradients of F_d^3 with F_d^4, both of them exhibit consistently high responses on edges, corners, and other features. It is speculated that F_d^4 learns from the gradient of F_d^3 with regional pooling. However, faint feature responses are inevitably present in regions where keypoints are invalid, such as lawns and lakes. It highlights the crucial role of the regional reliability weight W_s in inference.

3) Discussion on Reliable Weight: Boundaries, edges, and blocks are spaces with a universal distribution of keypoints, which are considered reliable in W_s. Conversely, the lawn and lake are given a lower response. Among all training modes, no significant changes are observed, indicating that the incorporation of keypoint-related modules does not affect the overall network's understanding of trustworthy regions. It is worth noting that in the case (a) of Fig. 13, despite being surrounded by the lawn, the inner house is not ignored and assigned high confidence in W_s. The detailed reliability of the TSK detector in regional inference is further proved.

4) Discussion on Score Map: Keypoints are determined with the highest features response within each grid of score map S. With the improvement of model integrity, significant differences are reflected in luminance between the keypoint-related regions and the regions without explicit

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semantics. It serves as evidence that the TSK network has developed a precise understanding of keypoints.

VI. Conclusion

In this article, a novel detector TSK is designed for trustworthy keypoint extraction. First, the mathematical model of keypoints is constructed based on representational and repeatability. On this basis, the multigranularity spaces, namely, pixel window, neighbor window, and existence entity, are set up. Meanwhile, a lightweight end-to-end network based on the codec structure and a saliency keypoint localization mechanism are designed for the keypoint location and descriptor extraction. To leverage keypoint attributes for feature refinement at different spatial scales, we construct three loss limitation modules: the existence reinforcement module, the trustworthy keypoint induction module, and the descriptor alignment module. Extensive experiments are carried out on four RSI datasets, and the performances for keypoint detection are verified by subsequent matching and registration tasks. Furthermore, with the assistance of feature maps and their corresponding gradient images, the keypoint recognition of the core network layers is represented through visualization, confirming that the keypoints generated by TSK are well-founded and trustworthy. By taking all components together, the TSK detector outperforms the state-of-art keypoint detectors. Moving forward, our research aims to investigate the synergistic relationship between detection and matching, uncovering the iterative interaction and collaboration between them during feature refinement.

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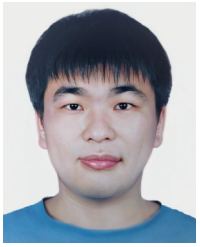
Jingyi Cao (Student Member, IEEE) received the B.E. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in 2019, where she is currently pursuing the Ph.D. degree in information and communication engineering.

Her research interests include remote sensing image intelligent processing, change detection, and 3-D processing of remote sensing data.



Yanan You (Member, IEEE) received the Ph.D. degree from the School of Electronic and Information Engineering, Beihang University, Beijing, China, in 2015.

He held a post-doctoral position at Beihang University from 2015 to 2017. He is currently an Associate Professor with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing. His research interests are in remote sensing image processing, deep learning, imaging detection, and intelligent perception.



Chao Li received the M.S. degree from the Beijing University of Posts and Telecommunications, Beijing, China, in 2023.

His research interest is in remote sensing image processing, especially in image registration and local descriptor construction.



Jun Liu (Member, IEEE) received the B.E. and Ph.D. degrees from the Department of Information Engineering, Beijing University of Posts and Telecommunications, Beijing, China, in 1998 and 2003, respectively.

He is currently the Director of the Research Center for Data Science, School of Artificial Intelligence, Beijing University of Posts and Telecommunications. His research interests include telecommunication big data analysis and network management.

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